From Answer Generators to Thinking Partners: Scaffolded LLM Tutors in Statistics Education

Abstract

Large language models (LLMs) such as ChatGPT can support statistics learning when their use is structured around pedagogy rather than answer provision. This paper introduces SMARTS, a reproducible six-phase scaffold for student–LLM interaction (State, Map, Ask, Reason, Think, Summarize) whose opening phase incorporates FRAME, a five-component checklist (Facts, Research objective, Assumptions, Method, Exact sticking point) that trains students to formulate precise statistical queries. We implement the protocol through course-aligned Custom GPT assistants (introductory and advanced) whose instructions constrain solutions, prefer hints, and draw on syllabus-consistent materials. The study reports deployment across two undergraduate courses at a Hispanic-Serving Institution and a three-wave attitude survey ($N = 32$ unique respondents; Course A $n = 14$, Course B $n = 18$; paired T1 $\rightarrow$ T3 subsamples $n = 13$ and $n = 16$). In the introductory course, paired trajectories on three items most directly tied to the protocol's intent—feeling prepared for lecture, recalling and applying prior concepts, and understanding lecture examples—show directionally clear positive shifts whose rank-biserial effect-size confidence intervals exclude zero (r_{rb} up to $+0.70$). The equity item declined in the introductory course ($r_{rb} = -0.27$; non-Hispanic students drove the decline in a pre-specified subgroup analysis) and the advanced course showed weaker, more heterogeneous paired effects overall. We report effect sizes with bootstrap confidence intervals rather than treating the small sample as powered for inference. We specify and release a rubric adapted from the SOLO taxonomy for auditing the depth of statistical reasoning exhibited in student–LLM transcripts; preliminary instructor-conducted audits document heterogeneous reasoning depth and illustrate how the assistants' graduated refusal-and-escalation logic redirects answer-extraction requests toward stepwise reasoning. All prompts, escalation clauses, course knowledge packs, survey instrument, coding rubric, and reproduction code are released as open educational resources. Results suggest that structured, course-aligned LLM use is teachable, auditable, and adaptable across course levels, supporting inquiry-based statistics instruction while upholding integrity and privacy.

Keywords: statistical reasoning, prompt engineering, cognitive apprenticeship, AI-assisted instruction, inquiry-based learning, SOLO taxonomy

1 Introduction

The ASA's Guidelines for Assessment and Instruction in Statistics Education recommend that introductory courses teach statistical thinking as an investigative process, foster active learning, and use technology to explore concepts and analyze data (GAISE College Report ASA Revision Committee, 2016). These recommendations have shaped a generation of curriculum reform. Yet the rapid adoption of large language models (LLMs) such as ChatGPT presents a new challenge: students now have instant access to tools that can execute an entire statistical analysis on their behalf. When a student pastes a homework question into an LLM and receives a complete solution, every phase of statistical thinking that Wild and Pfannkuch (1999) identified, from problem recognition through model reasoning to interpretation, is performed by the model rather than by the student. The GAISE vision of students as active investigators is precisely what is at risk.

Early responses to this challenge have been largely cautionary. Concerns about academic integrity and the erosion of independent thinking are well documented, both in commentaries emphasizing the disruptive potential of ChatGPT for higher education (Cotton et al., 2023; Rudolph et al., 2023) and in empirical evidence linking heavier AI tool use to reduced critical-thinking performance, mediated by cognitive offloading (Gerlich, 2025). Empirical evidence on how LLMs interact with statistical learning is only beginning to emerge. Ellis and Slade (2023), writing in this journal, offered an early exploration of ChatGPT's potential for statistics and data science education, drawing a parallel to the calculator debates of earlier decades (Ellington, 2003) and suggesting that guided use could yield benefits. More recently, Rao et al. (2026) found through task-based interviews with undergraduate students using Rtutor.AI that students' prior statistical knowledge influenced their ability to interpret and contextualize the tool's output, and that the tool did not fully absolve students of the need to think statistically. Their finding points to a critical gap: if the benefits of AI-assisted learning accrue mainly to the already-prepared, then unstructured LLM use may widen rather than narrow the disparities that statistics educators seek to address. This concern is particularly salient for heterogeneous student populations in institutions serving historically underserved learners, where equity considerations reinforce the rationale for structured pedagogical integration of AI tools rather than

unguided access (du Plooy et al., 2024).

This paper addresses that gap by specifying and evaluating a structured approach to student–LLM interaction designed for statistics courses. The approach has three components.

First, we introduce SMARTS (State, Map, Ask, Reason, Think, Summarize), a six-phase interaction protocol that structures how a student engages with an LLM when working on a statistical task. SMARTS is not a software product; it is a teachable, auditable scaffold that can be implemented in any LLM interface. Its phases operationalize principles from cognitive apprenticeship (Collins et al., 1989) and scaffolded instruction (Vygotsky, 1978; Wood et al., 1976): students must restate the problem in their own words, outline a statistical approach before computing, request only targeted resources, reason through one step at a time, reflect on assumptions and alternatives, and produce a summary linking results to course concepts. Each phase is designed to keep the student, not the LLM, as the agent of statistical reasoning, aligning with the GAISE emphasis on teaching statistics as an investigative process. The opening phase of SMARTS includes a structured prompt-construction checklist called FRAME (Facts, Research objective, Assumptions, Method, Exact sticking point) that teaches students to formulate precise, well-formed queries for statistical tasks.

Second, we describe two course-aligned LLM assistants, built as Custom GPTs within an institutionally licensed ChatGPT environment, that encode the SMARTS protocol in their system instructions. One assistant serves an introductory statistics course covering descriptive statistics, sampling, the normal distribution, the Central Limit Theorem, confidence intervals, hypothesis testing, chi-square tests, ANOVA, and simple linear regression, with all computational examples in R using the `mosaic` and `BSDA` packages. The other serves an advanced mathematical statistics course focused on point estimation theory, including efficiency, consistency, sufficiency, the Rao–Blackwell theorem, method of moments, and maximum likelihood estimation. Both assistants are constrained to instructor-uploaded syllabus materials, refuse to provide complete solutions, and privilege hints and guiding questions. This knowledge alignment, binding the assistant to the instructor's notation, methods, and datasets, addresses the well-documented problem of generic LLMs producing

off-syllabus or inconsistent statistical explanations (Ellis and Slade, 2023). All assistant prompts, system instructions, escalation clauses, and course knowledge configurations are released as open educational resources through an anonymized repository (see Section 8.7) to support adoption and adaptation.

Third, we report a classroom evaluation of SMARTS deployed across both courses during a single semester at a public, primarily undergraduate university designated as a Hispanic-Serving Institution. The evaluation draws on three data sources: (a) a three-wave Likert-scale attitude survey administered to consenting students at early, mid, and late points in the semester; (b) thematic analysis of open-ended survey responses; and (c) instructor-conducted transcript audits in which student–LLM conversations were coded using a rubric adapted from the SOLO taxonomy (Biggs and Collis, 1982) to assess depth of statistical reasoning.

The paper contributes to the emerging literature on AI in statistics education in three ways. It provides SMARTS, the first published interaction protocol designed specifically for student–LLM dialogue in statistics courses, grounded in established pedagogical theory and aligned with GAISE recommendations. It shows that Custom GPTs can implement pedagogical constraints with sufficient fidelity to support guided inquiry in both introductory and proof-based statistics. And it offers multi-source classroom evidence, combining attitudinal data with direct analysis of interaction quality, that structured LLM use is compatible with the investigative, process-oriented statistics instruction that the profession has long advocated.

The remainder of the paper is organized as follows. Section 2 reviews related work on LLMs in education, scaffolded instruction, and statistical thinking. Section 3 describes the two courses, their student populations, and the pedagogical context in which SMARTS was deployed. Section 4 presents the SMARTS protocol (including the FRAME checklist for precise prompting) and the design of the course-aligned assistants, with annotated examples of student–LLM interactions on statistical tasks. Section 5 describes the academic integrity and assessment framework. Section 6 presents the study's research questions, the survey instrument, the transcript audit procedure, and the analysis plan. Section 7 reports findings. Section 8 discusses implications, limitations, and directions for future work.

2 Related Work

2.1 Statistical Thinking and How Students Learn Statistics

Statistics education research distinguishes among statistical literacy (reading and interpreting statistical information), statistical reasoning (connecting concepts in the process of data analysis), and statistical thinking (the broader habit of mind that considers variation, context, and the investigative process as a whole) (Ben-Zvi and Garfield, 2004). Wild and Pfannkuch (1999) proposed a four-dimensional framework for statistical thinking in empirical enquiry comprising an investigative cycle (Problem, Plan, Data, Analysis, Conclusions), an interrogative cycle of generate–seek–interpret–criticize–judge, fundamental types of thinking (recognition of the need for data, transnumeration, reasoning with models, integrating context and statistics, considering variation), and dispositions such as skepticism and curiosity. Their framework emphasizes that statistical thinking involves not just executing procedures but continuously cycling between context and statistical models, questioning assumptions, and reflecting on the adequacy of conclusions.

The GAISE College Report translated these ideas into six curricular recommendations: teach statistical thinking, focus on conceptual understanding, integrate real data with context and purpose, foster active learning, use technology to explore concepts and analyze data, and use assessments to improve and evaluate student learning (GAISE College Report ASA Revision Committee, 2016). The GAISE recommendations have guided introductory-level course reform for nearly two decades and, as the report itself notes, apply at all levels of undergraduate statistics instruction.

Two features of these frameworks are especially relevant here. First, both emphasize that statistical reasoning is a *process*, not a product. Wild and Pfannkuch's investigative cycle is iterative: the student formulates, plans, collects, analyzes, and concludes, often looping back. Second, both call for active engagement. The GAISE report states that students should not be passive recipients of delivered solutions; they should be doing statistics, making decisions, and reflecting on what their analyses mean. Any technology introduced into a statistics course should support this active, process-oriented engagement, or at minimum not undermine it.

2.2 LLMs in Education and in Statistics

The release of ChatGPT in late 2022 prompted rapid commentary on both the risks and opportunities of large language models in education. Early concerns centered on academic integrity and the potential for students to outsource cognitive work to AI tools (Cotton et al., 2023; Rudolph et al., 2023). Within statistics education specifically, Ellis and Slade (2023) offered a constructive reframe in this journal, drawing a parallel to the calculator debates of the previous century (Ellington, 2003) and arguing that, as with calculators, the question is not whether to use the tool but how to integrate it so that it supports rather than supplants learning. They provided examples of ChatGPT aiding course material development and identified prompt design as a key lever for educational quality.

Since then, a small but growing empirical literature has examined how students interact with generative AI in statistics contexts. Rao et al. (2026), also in this journal, conducted task-based interviews with undergraduates using a generative AI coding tool and found that it could facilitate statistical thinking, particularly when students already possessed strong conceptual understanding. Critically, students with weaker foundations tended to accept AI-generated outputs with less interrogation, a finding consistent with broader concerns about over-reliance and shallow engagement in AI-assisted learning (Lee and Palmer, 2025). This result implies that the benefits of AI-assisted statistics learning may be unevenly distributed unless the interaction is structured to require the kind of active reasoning that GAISE and Wild and Pfannkuch's framework describe.

Beyond statistics, a broader literature on LLMs in education has identified scaffolded prompting as a key mechanism for promoting deeper engagement. Lee and Palmer (2025) conducted a systematic review of prompt engineering in higher education and found that prompts incorporating task restatement, concept activation, approach planning, and hint-seeking rather than solution-requesting promote metacognition and reduce passive reliance on AI output. These findings align with well-established principles from cognitive apprenticeship (Collins et al., 1989), in which expert guidance progresses through modeling, coaching, scaffolding with gradual release, articulation, reflection, and exploration, and with Vygotsky's (1978) concept of the zone of proximal development, where a more knowledgeable partner supports the learner just beyond independent capability. Wood et al.

(1976) formalized this support structure as scaffolding: temporary, responsive assistance that is withdrawn as competence develops.

Several recent studies have explored how these scaffolding principles can be operationalized in LLM-mediated instruction. In physics education, Krupp et al. (2024) compared undergraduate problem solving with and without ChatGPT access and found that students defaulted to copy-and-paste strategies, accepting roughly half of ChatGPT's answers as correct when they were not. In graduate robotics education, Brender et al. (2026) evaluated a structured prompting interface that required students to categorize and articulate prompts before submission, and found that students using the structured interface produced clearer task-specific prompts and that clearer prompts predicted higher learning gains. These findings collectively suggest that without deliberate interface or protocol design, unstructured LLM use in technical disciplines tends toward shallow interaction, while lightweight structural scaffolds can shift prompting behavior in pedagogically productive directions. However, this work has primarily targeted programming and engineering education, not statistics courses specifically. The dual demands of statistics education, simultaneously building conceptual understanding of statistical principles and developing computational fluency in tools such as R, create a distinctive instructional challenge that generic AI scaffolding frameworks do not fully address.

2.3 The Gap

Taken together, the existing literature establishes three things. First, effective statistics learning requires active, process-oriented engagement with the investigative cycle, not passive consumption of answers (Wild and Pfannkuch, 1999; GAISE College Report ASA Revision Committee, 2016). Second, unstructured student use of LLMs tends toward answer extraction and shallow engagement, with benefits concentrated among students who already have strong foundations (Rao et al., 2026; Krupp et al., 2024; Lee et al., 2025). Third, scaffolded prompting and cognitive apprenticeship principles offer a promising path for structuring LLM interactions (Brender et al., 2026), but no published work provides a reproducible, statistics-specific interaction protocol that students can learn and that instructors can audit, combined with classroom evidence of its effects.

This paper addresses that gap. We introduce SMARTS, a six-phase protocol for student–LLM dialogue grounded in cognitive apprenticeship and aligned with Wild and Pfannkuch's investigative and interrogative cycles. We describe how the protocol was implemented through course-aligned Custom GPTs in two statistics courses spanning introductory and advanced levels. And we report multi-source evidence from a semester-long deployment, combining attitudinal survey data with transcript-level analysis of interaction quality.

3 Course Context and Student Population

The study was conducted in two undergraduate statistics courses at a public, primarily undergraduate university in the western United States designated as a Hispanic-Serving Institution. Both courses were taught face-to-face by the same instructor during the spring 2025 semester. The courses differ substantially in level, content, student population, and the role that computation plays, providing a natural contrast for evaluating whether a single scaffolding framework can function across instructional contexts.

3.1 Course A: Introductory Statistics

Course A is an introductory statistics course that serves primarily as a required quantitative reasoning course for non-majors. The prerequisite is demonstrated basic mathematical readiness (a qualifying score on a placement exam or completion of an intermediate-algebra course); no prior statistics or programming experience is assumed. In the semester of the study, 14 students were enrolled. Thirteen were business administration majors across concentrations including marketing, accounting, finance, management, human resource management, and supply chain logistics. One student was a mathematics major on the statistics track. This enrollment profile is typical of the course, which draws broadly from programs that require statistical literacy but do not emphasize mathematical formalism.

The course uses an open-access textbook (Diez et al., 2014) supplemented by the instructor's lecture notes. Content follows a standard introductory sequence aligned with GAISE recommendations (GAISE College Report ASA Revision Committee, 2016): data

basics and collection principles, sampling strategies, observational studies and experimental design, descriptive statistics for numerical and categorical data, the normal distribution, the Central Limit Theorem, confidence intervals and hypothesis tests for proportions and means, chi-square tests, one-way analysis of variance, and simple linear regression, with a brief introduction to multiple regression. Instruction emphasizes interpretation and decision rules before computation, consistent with the GAISE emphasis on conceptual understanding and statistical thinking over procedural fluency.

The open-source statistical environment R (via RStudio) is used throughout the course for data exploration and analysis, with the `mosaic` and `BSDA` packages as the primary computational tools. For most students, this course is their first exposure to programming of any kind, creating a dual learning demand: acquiring statistical concepts while simultaneously developing basic computational skills (Wilcox and Lionelle, 2018; Hoffman and Elmi, 2021).

Assessment includes weekly quizzes administered through the learning management system (with multiple attempts to encourage mastery), five lab assignments involving data analysis in R, timed in-class quizzes with a revision-for-partial-credit policy that rewards correcting errors with shown work, an individual project and a group project both requiring data collection and analysis, and two closed-book in-person examinations. Students were permitted and encouraged to use the course-aligned LLM assistant (described in Section 4) for lab assignments, projects, R troubleshooting, and exam review; the assistant was not available during examinations.

3.2 Course B: Advanced Mathematical Statistics

Course B is an upper-division mathematical statistics course required for all tracks within the mathematics major (statistics, applied mathematics, and teaching). The prerequisite is a semester-long course in probability theory covering probability axioms, random variables, expectation, moment-generating functions, and common parametric families. In the semester of the study, 17 students were enrolled, all mathematics majors: four on the statistics track, two in applied mathematics, and eleven on the mathematics teaching track. The predominance of future mathematics teachers in this course is noteworthy: these students

will soon design their own instructional experiences, and their exposure to structured AI use in a rigorous mathematical context carries implications beyond their own learning.

The textbook is Wackerly et al. (2008). The course covers the theory of statistical inference: sampling distributions (t, chi-squared, F); point estimation including sufficiency, efficiency, consistency, the Rao–Blackwell theorem, method of moments, and maximum likelihood estimation; hypothesis testing and the likelihood ratio test; confidence intervals; and an introduction to Bayesian methods including prior and posterior distributions and credible intervals. Unlike Course A, the emphasis is on derivation and proof rather than applied data analysis. Students are expected to construct mathematical arguments, manipulate distributions algebraically, and evaluate the theoretical properties of statistical procedures.

R is used in this course for simulation and visualization of theoretical concepts (e.g., sampling distributions, Monte Carlo estimation of coverage probabilities) rather than for applied data analysis. Assessment consists of weekly problem sets requiring formal derivations, three closed-book in-person examinations, and a mandatory attendance component that includes in-class exercises. Students were permitted and encouraged to use the course-aligned LLM assistant for homework and exam review; the assistant was configured to surface relevant definitions and theorems from the instructor's lecture notes, ask the student to identify the next proof step, and reveal at most one step when the student was stuck, declining to provide complete proofs.

3.3 Integration of SMARTS and the Course-Aligned Assistants

In both courses, the SMARTS protocol and the Custom GPT assistants were introduced early in the semester as part of the regular instructional workflow, not as a separate experimental intervention. Before introducing the assistants, the instructor devoted class time to an AI literacy session covering the basics of how LLMs work, their capabilities and limitations, and the distinction between using AI for answer extraction versus using AI as a thinking partner. Figure 1 shows the input–output diagram used during this session to orient students to the technology. This preparatory discussion established norms for responsible use before any AI-assisted assignments were given.

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Figure 1: Overview of LLM input–output structure, used during the AI literacy session to orient students to how prompts shape model responses before introducing the SMARTS protocol.

Following the AI literacy session, the instructor introduced the SMARTS protocol, demonstrating the difference between unstructured and structured LLM interaction using a worked example relevant to each course. A one-page reference card summarizing the six SMARTS phases and the FRAME checklist was distributed. Students accessed the Custom GPT assistants through the institutionally licensed ChatGPT environment provided by the university system, which covers student and faculty data under existing institutional agreements and does not use interaction data for model training.

The syllabus for both courses included an explicit AI use policy stating that ChatGPT would be actively used for coding support (Course A) and conceptual exploration (Course B), with the expectation that AI interactions would supplement rather than replace the student's own reasoning. For any assignment in which AI was used, students were required to submit both their final solution and a transcript of the AI interaction (via share link or exported PDF), along with a brief reflection on how the interaction shaped their work. This dual submission served as the basis for the transcript audits described in

Section 6.

4 The SMARTS Framework and Course-Aligned Assistants

4.1 The SMARTS Protocol

Figure 2 summarizes the SMARTS pedagogical framework. The instructor configures a system prompt encoding the protocol and loads course-aligned reference material; the student engages the assistant through a gated FRAME opening followed by the six-phase SMARTS sequence; the assistant operates under a single cross-cutting content rule that freely shares general statistical content while withholding its application to the student's specific problem; and the instructor audits student transcripts to iterate on the configuration. The remainder of this section describes each element of the framework in detail.

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Figure 2: The SMARTS pedagogical framework. The instructor configures the system prompt, loads course-aligned knowledge, and audits student transcripts against a SOLO-based reasoning rubric (top). The student engages the assistant through a FRAME-gated opening followed by the six-phase SMARTS sequence (middle): State, Map, Ask, Reason, Think, Summarize. Throughout the interaction, the assistant freely shares general concepts, definitions, theorems, and standard techniques (SHARE column) but withholds their application to the student's specific problem (WITHHOLD column). Dashed arrows indicate the feedback loop: the student submits the transcript for instructor audit, and the instructor iterates the system prompt in response. The intended outcome is student reasoning made visible and auditable via the transcript record (bottom right).

SMARTS is a six-phase interaction protocol that structures how a student engages with an LLM when working on a statistical task. The acronym stands for State, Map, Ask, Reason, Think, and Summarize. Each phase is designed to keep the student as the active agent of reasoning while the LLM functions as a tutor that guides, checks, and prompts rather than delivers solutions.

Phase 1: State and clarify (FRAME). The student restates the statistical task in their own words. When needed, the assistant paraphrases to confirm mutual understanding and requests any missing information. This phase corresponds to the Problem stage of Wild and Pfannkuch's (1999) investigative cycle and to the articulation component of cognitive apprenticeship (Collins et al., 1989): making the student's understanding visible before any work begins.

Because the quality of the opening message largely determines the quality of the entire interaction, we equip students with FRAME, a five-component checklist for constructing a precise initial prompt. FRAME stands for:

F – ***Facts.*** State what you have. In an applied setting: sample size, variable names and types, data collection method, and relevant summary statistics. In a proof-based setting: the random variables, distributional family, and given quantities.

R – ***Research objective.*** State in one sentence what you are trying to accomplish: estimate a parameter, test a hypothesis, compare groups, build a model, or prove a property such as unbiasedness or sufficiency.

A – ***Assumptions.*** State the conditions you believe hold or need to verify: normality, independence, equal variances, known versus unknown σ, random sampling. For proofs: the regularity conditions and theorems you believe are available.

M – ***Method.*** State which procedure or proof strategy you think applies and why. Committing to a method before receiving AI feedback forces the student to reason about the mapping from problem to procedure, a central element of statistical thinking (Wild and Pfannkuch, 1999).

E – ***Exact sticking point.*** Identify precisely where you are stuck. "I do not understand" is not actionable. "I computed $t = 1.87$ with $df = 29$ but am unsure how to find the p-value from the t-table" is immediately useful.

FRAME teaches students that a precise prompt is not a technical trick for getting better AI output; it is an act of statistical problem formulation. Each component maps to a recognized element of statistical thinking: Facts to context and data, Research objective to

problem definition, Assumptions to model reasoning, Method to procedural selection, and Exact sticking point to the interrogative cycle of questioning and self-monitoring (Wild and Pfannkuch, 1999). In class, the instructor demonstrated the difference between a vague one-line request and the same question posed with all five FRAME components, showing how the LLM's response changed in specificity, accuracy, and alignment with course materials. The FRAME checklist was printed on the reverse side of the SMARTS reference card distributed to students; both are available in the supplementary repository.

Phase 2: Map a plan. Before any calculation or code, the student outlines an approach: identifying the parameter of interest, choosing a statistical procedure, specifying assumptions, and anticipating the form of the conclusion. The assistant critiques the plan rather than supplying one. This phase parallels the Plan stage of the PPDAC cycle (Wild and Pfannkuch, 1999) and serves as scaffolding (Wood et al., 1976) that supports planning while leaving the intellectual work with the student.

Phase 3: Ask and gather. The assistant surfaces only the minimal resources required at that moment: a relevant definition, formula, theorem, or section of the course notes. Course-specific materials uploaded to the assistant are prioritized over open-ended web knowledge, ensuring that explanations match the instructor's notation and methods.

Phase 4: Reason step by step. The student executes one step at a time and receives brief checks or targeted hints. The assistant withholds complete solutions, escalating to reveal a single next step only after sustained impasse. This phase operationalizes the coaching and scaffolding with gradual release components of cognitive apprenticeship and keeps the student within their zone of proximal development (Vygotsky, 1978).

Phase 5: Think critically and iterate. The student reflects on alternative approaches, robustness to assumptions, and potential sources of error. The assistant prompts self-explanation rather than providing conclusions. This phase targets the interrogative cycle that Wild and Pfannkuch (1999) identified as central to statistical thinking: the generate–seek–interpret–criticize–judge loop that distinguishes deep engagement from mechanical

execution.

Phase 6: Summarize and verify. The assistant closes with a succinct recap of what was asked, what was done, and why the result is appropriate, linking back to course concepts and noting limitations. This phase reinforces the reflection and exploration stages of cognitive apprenticeship and supports the Conclusions stage of the investigative cycle.

The protocol is deliberately general: it applies to any statistical task, from computing a confidence interval to constructing a maximum likelihood proof. Its value lies not in prescribing what statistical content to cover but in structuring the dialogue so that the student must think at each stage rather than passively receive. In practice, SMARTS is implemented through system instructions embedded in the Custom GPT assistants described below; students encounter the protocol through the assistant's behavior without needing to memorize it.

To make the protocol concrete and immediately adoptable, we reproduce the domain-neutral skeleton prompt in Figures 3 and 4 (two parts comprising a single system prompt). Any instructor can paste this text as a system prompt or custom instruction into ChatGPT, Claude, Gemini, or a comparable interface and obtain an assistant that enforces the FRAME opener, the six-phase sequence, and the boundary between *general content* (freely shared) and *application to the student's specific problem* (withheld). Course-aligned specializations that add course-specific notation, textbook references, and computational constraints are described in Section 4.2 and released in full in the supplementary repository (Section 8.7). The skeleton prompt is released under a Creative Commons Attribution (CC BY 4.0) license with the citation request noted in the figure captions.

Figure 3: The discipline-neutral SMARTS thinking-partner system prompt, part 1: FRAME opening requirement and SMARTS phase instructions. Reproduced verbatim as deployed. The complete prompt continues in Figure 4; both parts together constitute a single system prompt that an instructor pastes into ChatGPT, Claude, Gemini, or a comparable interface. Released under a Creative Commons Attribution (CC BY 4.0) license; reuse in any context is permitted provided this paper is cited.

```
I want you to adopt the role of a thinking-partner tutor for a
problem or task I am working on, using the SMARTS framework. You
will not hand me a full solution. You coach me one step at a time.

Before engaging with the problem, require me to give you a FRAME
opening:
  F -- Facts: what I have -- the data, evidence, given quantities,
      source material, inputs, or context.
  R -- Research objective: one sentence stating what I am trying
      to accomplish -- prove, analyze, decide, build, write,
      design, compute, argue, interpret, or create.
  A -- Assumptions: the conditions, premises, frameworks,
      principles, theories, or methods I am taking as given.
  M -- Method: the approach, procedure, or strategy I think
      applies, and why.
  E -- Exact sticking point: the specific step or concept where
      I am stuck.
If my FRAME is incomplete or vague, ask for the missing piece. Do
not proceed until FRAME is adequate.

Then work through the SMARTS phases with me. Announce the phase
briefly at the start of each reply.
  S -- State: confirm my setup.
  M -- Map: ask me to sketch a plan. Critique it. Do not supply
      one.
  A -- Ask: give me only the minimum I need right now -- one
      concept, one definition, one example, one technique at a
      time. No solution dumps.
  R -- Reason: I do each step; you check. If a step is wrong,
      name the step to re-examine. Do not give me the corrected
      version.
  T -- Think: prompt me to self-explain. Ask why an approach
      applies, what its conditions require, what would break
      without them.
  S -- Summarize: after I finish, ask me to recap what I did and
      why, then reinforce the key ideas.
```

Figure 4: The discipline-neutral SMARTS thinking-partner system prompt, part 2: core content boundary and closing instructions. Continued from Figure 3. Course-aligned specializations (Course A introductory statistics R-tutor, Course B mathematical statistics proof coach), assessment-delivery variants (Mini-Gauntlet family), and chapter-specific specializations are released in the supplementary repository (Section 8.7).

```
Core rule on content:
  FREELY SHARE general content -- concepts, definitions,
  principles, frameworks, theorems, standard techniques,
  historical context, and illustrative examples drawn from
  unrelated contexts -- as clean explanatory material.
  DO NOT apply that content to my specific problem. Do not write
  my essay, draft my argument, complete my code, compute my
  answer, produce my analysis, make my decision, design my
  solution, or otherwise finish the work on my behalf. The
  application is mine.

If I ask for the full solution, decline and redirect to the next
step. If I am stuck after a genuine attempt, reveal only the
first step to get me unstuck, then stop and wait.

Keep replies short. No emojis. No filler. If I drift off the
task, tell me briefly and redirect.

Now wait for my FRAME.
```

4.2 Course-Aligned Custom GPT Implementation

The SMARTS protocol was operationalized through two Custom GPTs built within the university's institutionally licensed ChatGPT environment. Custom GPTs are versions of ChatGPT configured with persistent system instructions and curated reference materials; no programming is required. By embedding the SMARTS phases in the system instructions, the protocol governs every interaction by default, removing the burden on students to apply the framework manually. This approach also ensures consistency: unlike ad hoc prompting, which drifts over long conversations, the Custom GPT maintains its tutoring persona and refusal policies throughout a session.

Each assistant was configured with three core design principles.

Knowledge alignment. The assistant's reference knowledge was restricted to instructor-uploaded materials: lecture notes, worked examples, and approved problem sets. When a student asked a question, the assistant searched these materials first and drew on them preferentially. This binding of AI knowledge to the instructor's curriculum reduced off-syllabus explanations, notation inconsistencies, and hallucinated content, addressing a well-documented limitation of generic LLMs in educational settings (Ellis and Slade, 2023).

Solution refusal with graduated escalation. The assistant was instructed never to provide a complete solution in a single response. When a student requested one, the assistant acknowledged the request, gently redirected to the current step, and offered a guiding question. If the student remained stuck after multiple exchanges, the assistant revealed one additional step with reference to the course materials, then returned to guiding mode. This escalation structure implements scaffolding with gradual release: support is available but calibrated to the minimum necessary.

Scope restriction. Each assistant responded only to questions within the scope of its course. Off-topic queries received a polite refusal and redirection. This constraint prevented the assistant from functioning as a general-purpose chatbot and reinforced its role as a course-specific learning partner.

The two assistants differed in ways that reflected their courses' distinct pedagogical needs. The Course A (introductory) assistant assumed no prior coding experience, used the `mosaic` and `BSDA` packages exclusively unless alternatives were unavoidable, explained every line of R code it provided, and prioritized interpretation and decision rules before computation. The Course B (advanced) assistant drew exclusively on the instructor's lecture notes for point estimation theory, surfaced relevant lemmas and definitions when proof structure was requested, asked the student to identify the next logical step before providing guidance, and declined to supply complete proofs. Both assistants included a refusal clause for requests outside their respective course scopes and an escalation clause for persistent impasse. The full system instructions for both assistants are provided in the supplementary repository.

4.3 SMARTS in Action: Annotated Examples

To illustrate how the SMARTS protocol shapes student–LLM interaction on statistical tasks, we present four annotated excerpts from student transcripts submitted during the semester: two from Course A (introductory) and two from Course B (advanced). Each excerpt has been lightly edited to remove identifying information but otherwise reproduces the student–assistant exchange verbatim, including typographical features of student turns. SMARTS phase labels appear in brackets; we record the SOLO reasoning level of the student's highest-level turn with a one-sentence rubric justification. The full transcripts corresponding to all four excerpts, together with additional sessions spanning canonical course topics, appear in Appendix D.

We acknowledge up front that students in this study did not consistently apply the SMARTS phase structure to their own opening turns; many interactions began as simple requests for help or for a solution. This is expected, and it is the primary reason the protocol is encoded in the assistant's system instructions rather than left to student initiative. The excerpts below were therefore chosen not to display ideal SMARTS compliance, but to show how the assistant's scaffolded behavior responds to the range of openings students in fact produce.

Example 1: Course A — Hypothesis test on Lipitor efficacy claim (custom GPT). This excerpt is from a hypothesis-testing assignment in which the student evaluates a manufacturer's claim that a cholesterol medication reduces LDL cholesterol by a specified amount. The student opens by asking the assistant to work the problem; the assistant redirects to a FRAME-style decomposition, and the student proceeds to compute the test statistic and interpret the result in context.

> **[Phase S — State and clarify]**
>
> Student: "can u do this problem for me"
>
> Assistant: "I can't solve it for you, but I can coach you through it. Give me your FRAME: Facts (what data, sample size, standard deviation, claim value), Research objective (what are we testing), Assumptions (what conditions does the test need), Method (what test), Exact sticking point."

[Phase M — Map a plan]

Student: [provides FRAME with the 25-patient sample mean and SD, states $H_0 : \mu = 20$ vs. $H_a : \mu < 20$, names a one-sample t-test, asks how to handle the "one-sided" direction]

Assistant: [confirms the one-sample t-test is correct; asks the student to verify the three conditions in their own words before computing the statistic]

[Phase R — Reason step by step]

Student: "$t = (\bar{x} - 20)/(s/\sqrt{n})$, I got $t = -2.14$, $df = 24$"

Assistant: "Good. Now you need the p-value. What is the direction of the alternative, and what does that mean for how you read the t-table?"

[Phase S — Summarize and verify]

Student: "p-value about 0.021, so we reject H_0 at $\alpha = 0.05$. There is statistical evidence that mean LDL reduction is less than the 20 claimed by the manufacturer."

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the test statistic, degrees of freedom, one-sided p-value, significance decision, and context-appropriate conclusion into a coherent claim about the manufacturer's statement.

Example 2: Course A — ANOVA and Tukey HSD on GPA data (custom GPT). This excerpt is drawn from an assignment comparing mean GPAs across four universities. It illustrates how the assistant scaffolds both R code and conceptual reasoning within a single session, and how it enforces the distinction between procedural correctness and statistical understanding. After the student runs an ANOVA and reports a small p-value, the assistant refuses to move on to the post-hoc test until the student articulates why the post-hoc is needed.

[Phase T — Think critically and iterate]

Student: "got it pvalue is soooooo tiny we go to tukey? explainf irst"

Assistant: "Good — now slow down. You said: 'p-value is so tiny → we go to Tukey.' That's procedurally correct, but I want you to understand *why* before we move on. What does a very small p-value in the ANOVA F-test actually tell you about the null hypothesis? ... Why can't we just run a bunch of regular t-tests?"

[several turns on family-wise error rate; student counts 6 pairwise comparisons and articulates inflation of Type I error]

[Phase R — Reason step by step]

Student: [reports `TukeyHSD` output] "so except U3-U2 and U4-U3 everything else is dig different?"

Assistant: "Careful — you jumped too fast and made a mistake. A pair is significantly different if `p adj` < 0.05 *or* the CI does not include 0. Check U3 − U2: `p adj` $= 0.0117$, CI $= (0.03, 0.33)$. Is that significant?"

[Phase S — Summarize and verify]

Student: "so all are different?"

Assistant: "Yes. Now don't stop at 'all are different' — state the ordering of the means from lowest to highest and write a complete conclusion."

[Student: "ty"]

Assistant: "Before we close — you didn't finish. Just saying 'ty' skips the most important part of the assignment. Order the universities by mean GPA lowest to highest."

Student: "U1 < U2 < U3 < U4, all pairs statistically significantly different".[1]

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the ANOVA omnibus result, the multiple-comparisons rationale, the correct application of the Tukey decision rule across all six pairs, and the ordered conclusion into a coherent inferential statement.

Example 3: Course B — Unbiasedness of $\hat{\theta} = nY_{(1)}$ (custom GPT). This excerpt shows the "proof construction coaching" custom GPT, whose system instructions draw an explicit line between *general theorems, definitions, and formulas*, which the assistant will share freely, and *application to the student's specific problem*, which the student must

[1]Institution names in this transcript have been replaced with generic labels U1–U4 for blind review; the original exchange referenced four specific public universities, ordered in the student's conclusion from lowest to highest sample mean GPA.

complete. The example shows the assistant intervening when the student attempts to jump from the general minimum-density formula to the specific exponential substitution.

> **[Phase R — Reason step by step]**
>
> Student: "$f_{Y_{(1)}}(y) = n(1 - F(y))^{n-1} f(y) = n \cdot e^{-ny/\theta}$ right?"
>
> Assistant: "You're very close — but this is exactly where I need to stop you from jumping to the final expression. What you wrote, $f_{Y_{(1)}}(y) = n(1 - F(y))^{n-1} f(y)$, is correct as the general formula. But the step $= ne^{-ny/\theta}$ is the application to your specific distribution, and I'm not going to complete or confirm that for you. Instead, let's slow it down and make sure your substitution is solid. Write explicitly: What is $F(y)$? What is $1 - F(y)$? What is $f(y)$? Then plug those three pieces in."
>
> **[Phase R — Reason step by step, continued]**
>
> Student: "yeah i did that im miisng a $/\theta$"
>
> Assistant: "That sounds like the kind of algebra check you should make here. If $f(y)$ contains a factor $1/\theta$, that factor stays unless something else cancels it. Check: did $f(y)$ contribute a $1/\theta$? Did anything in $(1 - F(y))^{n-1}$ cancel that?"
>
> **[Phase S — Summarize and verify]**
>
> Student: "$E(Y_{(1)}) = \theta/n$, so $E(nY_{(1)}) = \theta$. its theta."
>
> Assistant: "Yes. That proves $\hat{\theta} = nY_{(1)}$ is unbiased. Big picture takeaway: use a general order statistic formula; substitute your distribution; recognize a known distribution; use a standard expectation result; apply linearity. That's the full pipeline you'll reuse on other problems."

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the order-statistic minimum-density formula, the specific exponential distribution, recognition of the exponential rate parameter, the standard expectation result, and linearity of expectation into a complete unbiasedness proof.

Example 4: Course B — Unbiasedness of S^2 with a student-written system prompt. This excerpt is drawn from a session in which the student did not use the course-aligned custom GPT but instead supplied their own system-prompt to ChatGPT

instructing it to behave as a step-by-step statistics tutor. The student's prompt is pedagogically reasonable but softer than the SMARTS specification: it asks for "hints, not entire solutions" but does not enforce a FRAME opener, does not prohibit completing the application after a student's request, and offers to "reveal only the first step and prompt me to proceed" when the student struggles. We include this example to illustrate what a well-intentioned but unstructured alternative looks like on an adjacent topic (unbiasedness of the sample variance), and to clarify by contrast what the SMARTS-governed assistant in Example 3 was enforcing.

> **[Phase M — Map a plan]**
>
> Student: "So I need to find the expectation of this statistic and show it is σ^2?"
>
> Assistant: "Exactly! Your goal is to compute $E[S^2]$ and show that $E[S^2] = \sigma^2$. Would you like to attempt computing $E[S^2]$ directly, or shall I suggest a helpful identity to simplify the process first?"
>
> **[Phase R — Reason step by step]**
>
> Student: "I'll just expand and see what happens"
>
> Assistant: "Perfect! That's a great next move. Try expanding that sum. You might want to write $(Y_i - \bar{Y})^2 = Y_i^2 - 2Y_i\bar{Y} + \bar{Y}^2$. Then plug that into the summation and simplify."
>
> **[Phase R — Reason step by step, continued]**
>
> Student: "$\sum Y_i^2 - 2n\bar{Y}^2 + n\bar{Y}^2 \ldots$ can take $n-1$ out now $\ldots$ $E(Y_i^2) = V(Y_i) + E(Y_i)^2$?"
>
> Assistant: "Excellent progress! Let's carefully walk through and confirm each of your thoughts. [*confirms all three steps explicitly, completes the identity* $E[Y_i^2] = \sigma^2 + \mu^2$] $\ldots$ You're almost there!"

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the sample-variance identity, linearity of expectation, the variance-mean relation $E[X^2] = \mathrm{Var}(X) + (E[X])^2$, and the $\mathrm{Var}(\bar{Y}) = \sigma^2/n$ result into an unbiasedness argument.

Examples 3 and 4 are illustrative of two different scaffolding regimes on adjacent topics. Both interactions reach the relational SOLO level, and both students produce a coherent unbiasedness argument. The interactions differ, however, in the assistant's posture at the

inflection points. The custom-GPT assistant in Example 3 refuses to confirm the student's specific substitution and routes the student back to the general formula for a self-diagnosed algebra check; it reveals a structural pipeline as a summary rather than a completed derivation. The student-prompted assistant in Example 4 confirms each step as it is produced, supplies the next identity when invited, and reaffirms partial progress ("great next move," "excellent progress," "you're almost there"). Both scaffolding styles can produce the relational outcome on a student with strong foundations; the design question is which style better serves students who do not yet have those foundations. This is precisely the question the current study is not powered to answer (Section 6.6), and that we identify in the Discussion as a first-priority direction for subsequent work (Section 8). The excerpts are reproduced here to make the scaffolding distinction concrete for readers; the full transcripts are in Appendix D.

These four examples are not intended as average cases. The full sampling rationale and distribution of coded reasoning levels are reported in Section 7.6.

5 Academic Integrity and Assessment Design

The integration of LLM tools into statistics coursework raises well-documented concerns about academic integrity, particularly the risk that students will use AI to extract answers rather than to learn (Cotton et al., 2023; Ellis and Slade, 2023). Our approach addresses these concerns not by restricting AI use but by making it transparent, auditable, and graded for process rather than product.

Four design principles guided the assessment framework in both courses.

First, AI use was explicitly permitted and structured rather than prohibited. The syllabus for each course stated that the Custom GPT assistant could be used for specified assignments (lab work, homework, exam review) but not during in-person examinations. By removing ambiguity about when AI is allowed, we reduced the incentive for covert use and reframed the assistant as a sanctioned learning tool.

Second, assessment emphasized process over product. For any assignment in which AI was used, students submitted two artifacts: their final solution with shown work, and the AI

interaction transcript (a share link or exported PDF from the chat platform), accompanied by a brief written reflection describing how the interaction shaped their understanding. This dual submission enabled the instructor to evaluate not only correctness but also the quality of the student's engagement with the assistant.

Third, transcript review focused on specific indicators of learning-oriented interaction: evidence of multi-turn dialogue rather than single-query answer extraction, instances where the student challenged or verified the assistant's suggestions, use of the stepwise scaffolding consistent with SMARTS rather than "solve this" prompts, and final explanations in the student's own words with appropriate justification. When transcripts revealed shallow engagement (e.g., restarting a conversation to bypass the scaffolding), the instructor addressed this through individual feedback rather than punitive measures; most students adjusted their behavior once the emphasis on demonstrated understanding was reinforced.

Fourth, in-person examinations served as a calibration check. Because examinations were closed-book and conducted without AI access, they provided an independent measure of whether students had internalized the statistical concepts practiced with the assistant. The coexistence of AI-assisted formative work and AI-free summative assessment created a natural accountability structure: students who relied on the assistant without genuine engagement would be unable to perform on examinations.

These principles draw on the broader pedagogical argument that, in professional settings where AI tools are increasingly ubiquitous, preparing students to use them with integrity and critical judgment is more valuable than creating artificial AI-free environments (Ellis and Slade, 2023).

6 Methods

6.1 Research Questions

The study addresses three research questions:

RQ1. How do students in introductory and advanced statistics courses perceive the usability, fairness, and educational value of course-aligned LLM assistants governed by the SMARTS protocol, and do these perceptions change over the course of a semester?

RQ2. What patterns characterize students' open-ended descriptions of their experiences with the assistants, and how do these patterns differ across course levels?

RQ3. To what extent do student–LLM interaction transcripts exhibit the stepwise reasoning and self-explanation targeted by SMARTS, as assessed by a rubric adapted from the SOLO taxonomy?

6.2 Study Design

This is a descriptive, multi-wave classroom study. The design is observational: the SMARTS protocol and Custom GPT assistants were integrated into regular coursework in both courses, and data were collected to characterize student attitudes and interaction quality over the semester. No control group was used, as the primary aim was to document what structured LLM use looks like in practice and how students respond, not to isolate a causal treatment effect. This design is appropriate for an early-stage intervention study in which the protocol is being specified and refined for the first time, and where randomization is impractical and ecological validity is paramount.

6.3 Participants and Ethical Approval

Human participants research was approved by the university's Institutional Review Board (protocol number anonymized for review). Informed consent was obtained from all participants. Fourteen students in Course A (Introductory Statistics) and eighteen in Course B (Advanced Mathematical Statistics) responded to at least one survey wave, for a total of $N = 32$ unique respondents across both courses. Wave-level response counts in the attitude survey were lower than the course total due to normal class absenteeism on survey administration days (reported in Section 6.5 and Tables 2 and 3). Only data from consenting participants are reported. Demographic characteristics are summarized in Table 1.

6.4 Data Sources

Three data sources were collected.

Likert-scale attitude survey. An 11-item instrument was administered at three time points: early semester (T1, approximately week 3), mid-semester (T2, approximately week 8), and late semester (T3, approximately week 14). Items assessed perceived support for learning, encouragement of critical thinking, usability and accessibility, helpfulness of feedback relative to the instructor, support for concept recall and application, support for learning regardless of background, and perceived preparation for lecture. Each item used a 5-point Likert-type response scale (1 = Strongly Disagree to 5 = Strongly Agree). The instrument was developed for this study and administered identically across both courses and all three waves.

Open-ended survey responses. In addition to the Likert items, the instrument included five open-ended prompts administered at every wave: (i) how AI tools help or hinder work–life–class balance; (ii) which features of the AI tutor the student found most helpful or in need of improvement; (iii) whether the AI tutor accommodates the needs of all students equally, and if not, what specific challenges or barriers the student or others had experienced; (iv) how using the AI tutor had affected the student's ability to think critically and solve problems independently, including specific instances; and (v) any suggestions for improving the AI tutoring systems. Each prompt was administered at T1, T2, and T3; because student response rates to the open-ended items were high (approximately 100% of attending students at each wave), the full corpus comprises 189 substantive responses from Course A students and 239 substantive responses from Course B students, for a total of 428 nonblank responses (5 prompts $\times$ 3 waves per respondent, counting any text entry; varies with attendance). Stricter content filters (e.g., excluding responses of three characters or fewer) yield a smaller effective corpus of 395 entries; we retain all 428 as the denominator because meaningfully short responses ("N/A," "none") are themselves interpretable signals.

Student–LLM interaction transcripts. For assignments where AI use was permitted, students submitted transcripts of their interactions with the Custom GPT assistants. These transcripts were collected as part of routine coursework and were available for analysis with participants' consent.

6.5 Analysis Plan

Quantitative analysis of Likert data. The survey administration retained a course-internal respondent identifier (distinct from institutional identifiers and not mappable to student records) which permits paired matching across waves. Given the small sample sizes ($n = 13$ paired in Course A and $n = 16$ paired in Course B; see Section 7.1), the ordinal response scale, and the within-subjects repeated-measures structure, we use nonparametric paired methods throughout. Treating ordinal Likert responses as metric is known to produce systematically biased estimates of both effect size and uncertainty (Liddell and Kruschke, 2018); we therefore avoid t-tests and Pearson-correlation-based effect sizes on the item responses.

For each item and course, we report descriptive statistics at T1 and T3 paired for the same students: wave means and standard deviations, the mean paired difference $\bar{\Delta} = T3 - T1$, and the tripartite count of students whose T3 response exceeded (+), matched (=), or fell below (−) their T1 response. The primary inferential test is the Wilcoxon signed-rank test with Pratt's method for zero differences (Pratt, 1959), using the exact null distribution throughout. Paired $n \leq 16$ falls well below the $n \approx 20$ threshold at which the asymptotic normal approximation becomes reliable (Bellera et al., 2010), making the exact method not merely preferred but required.

Effect sizes are reported as the matched-pairs rank-biserial correlation r_{rb} (Kerby, 2014), computed as the difference between the proportion of favorable and unfavorable signed ranks. The rank-biserial correlation is the recommended effect-size measure for the Wilcoxon signed-rank test (King et al., 2011; Tomczak and Tomczak, 2014) and ranges from -1 to $+1$ with interpretive benchmarks of $|r_{rb}| \geq 0.1$ (small), ≥ 0.3 (medium), and ≥ 0.5 (large). Ninety-five percent confidence intervals for r_{rb} are obtained from 5,000 percentile-bootstrap replications; we follow the conservative recommendation of Mangiafico (2016) to flag bootstrap intervals at extreme values or with small cell counts as potentially unreliable. Items whose r_{rb} 95% CI excludes zero are interpreted as having directionally clear effects; items whose CI spans zero are reported as inconclusive regardless of p value. This CI-based decision rule is the effect-size-first framing that Wilkinson and APA Task Force on Statistical Inference (1999) and subsequent APA guidelines advocate.

Subgroup analyses of trajectory differences. In addition to the primary item-level paired analyses, we report four pre-specified subgroup comparisons per course, each testing whether the T1 $\rightarrow$ T3 trajectory on a theoretically motivated item or composite differs between two demographic subgroups. The four pre-specified comparisons (Section 7.4) are (A1) Hispanic/Latino identity versus the Q7 equity item, (A2) prior AI tool use versus the composite mean of Q1–Q11, (A3) English as primary language versus the Q2 critical-thinking item, and (A4) employment hours (at/above versus below the observed median) versus the Q8 preparation-for- lecture item. For each comparison, we compute the T3−T1 difference score for each student and then apply an exact Mann–Whitney U test (Mann and Whitney, 1947) comparing difference scores between subgroups; Glass rank-biserial correlation serves as the effect size, with 95% CI by percentile bootstrap on 5,000 replications. All four analyses (eight including both courses) are reported regardless of p value or effect size magnitude to avoid selective reporting (Wilkinson and APA Task Force on Statistical Inference, 1999).

Three-wave sensitivity. Twenty-five of the 32 respondents (78.1%; Course A $n = 11$, Course B $n = 14$) provided complete data at all three waves (Figure 5), enabling two classical nonparametric repeated-measures sensitivity analyses on the complete-case subsample: the omnibus Friedman test and Page's L test for ordered trend (Hollander et al., 2014; Page, 1963). Page's L is the paired analog of Jonckheere–Terpstra and is the appropriate sensitivity cross-check for a three-wave attitude panel. Results (reported in Appendix B and in the supplementary repository) are directionally concordant with the primary paired T1 $\rightarrow$ T3 Wilcoxon analyses: no item reaches $p < 0.05$ under either test, and the same three Course A items (Q3 recall/apply, Q8 prep for lecture, Q9 understand examples) produce the smallest p-values and the largest Page's L statistics. A pooled ordered logistic regression on the full unbalanced panel (with wave as ordered numeric predictor) is reported in the same appendix as a parametric cross-check; the sign of its trend coefficient agrees with the paired T1 $\rightarrow$ T3 direction on 21 of the 22 item$\times$course combinations. We describe this agreement as parametric cross-check rather than methodological convergence because, under a true underlying monotone trend, the two procedures share non-centrality by construction; agreement contributes to robustness against parametric misspecification

but does not constitute independent corroboration.

Why not cumulative-link mixed models. A cumulative-link mixed model with random intercept per student (Christensen, 2015) would in principle be the parametric best-estimator for these data, modeling the within-subject correlation directly. At the available sample sizes, however, each per-item model attempts to estimate four ordinal thresholds, one slope, and one between-subject variance from roughly 33–48 wave-level observations arising from only 11 or 14 clusters. Simulation-based guidance (Bolker et al., 2009; McNeish and Kelley, 2019) recommends at least five observations per random-effect level with the additional expectation that between-cluster variance is well identified, which is pushed to its limit with three observations per cluster. We therefore report the exact paired Wilcoxon test as primary and the pooled ordered-logit model as parametric sensitivity, deferring cumulative-link mixed models to the supplementary repository as an optional robustness fit.

Inferential framing. Given the exploratory character of the study, the small sample sizes that render p-values unreliable as inferential instruments, and our effect-size-first decision rule, we do not apply multiple-comparison adjustments to the 22 item-level tests per course or the eight subgroup comparisons. Each Likert item corresponds to a distinct substantive construct (preparation, feedback, equity, and so on) rather than a disjunctive test of a single composite hypothesis, placing our inference in the individual-testing regime for which global alpha adjustment is not required (Rubin, 2021). Consistent with the American Statistical Association's post-2019 guidance (Wasserstein et al., 2019; Amrhein et al., 2019), we avoid the language of "statistical significance," treat confidence intervals as compatibility intervals rather than as accept/reject tests, and frame effect-size point estimates as directional summaries rather than discoveries. Readers are invited to interpret the pattern of findings holistically, weighting effect-size point estimates and their compatibility intervals above individual p-values. The exact Wilcoxon and Mann–Whitney procedures yield valid Type I error control at small n (Meek et al., 2007; Bellera et al., 2010); the conservative reporting choice is in the interpretation of borderline results, not in the testing procedure itself.

Missing data. Wave-level response counts vary across administrations due to normal

class absenteeism (Course A: $n = 14, 12, 13$ at T1, T2, T3; Course B: $n = 17, 14, 17$; paired n reported in Tables 2 and 3 varies slightly by item due to occasional per-item non-response). Analysis is complete-case within each paired comparison; we do not impute missing responses given the small sample sizes and the possibility that attendance patterns are substantively associated with attitudes toward the assistant. All analyses are conducted in Python and R; code is provided in the supplementary repository.

Qualitative analysis of open-ended responses. The 428 open-ended responses are analyzed thematically following the six-phase reflexive thematic analysis procedure of Braun and Clarke (2006): (i) familiarization through repeated reading of the full corpus by course; (ii) initial code generation; (iii) searching for themes by clustering codes; (iv) reviewing themes against the coded extracts and the full dataset; (v) defining and naming themes; and (vi) the present report. We use a hybrid inductive–deductive approach: deductive scaffolding comes from the five survey-prompt domains (balance, features and improvements, equity, critical thinking, suggestions), and inductive codes are generated within each domain from the response data. Per the study design, we conduct separate thematic analyses for Course A and Course B rather than merging the corpora, reporting themes side by side and noting convergences in the Discussion rather than in the Results. Coding is conducted by a single coder (the instructor of both courses), with the reflexivity considerations described in Section 6.6. The full codebook, all coded extracts with code assignments and coder notes, and the de-identified essay corpus are released in the supplementary repository (Section 8.7) to support independent verification of theme attributions and to enable future multi-coder reliability studies.

Transcript audit using SOLO-adapted rubric. A purposive sample of student–LLM transcripts is coded using a rubric adapted from the SOLO taxonomy (Biggs and Collis, 1982; Brabrand and Dahl, 2009). The rubric classifies the highest level of statistical reasoning exhibited by the student across the interaction:

Prestructural. The student's contributions are irrelevant to the statistical task or consist entirely of pasting the problem without engagement.

Unistructural. The student addresses one aspect of the task with minimal elaboration (e.g., identifies the correct test but cannot articulate why).

Multistructural. The student addresses several aspects independently without integration (e.g., specifies hypotheses and computes a test statistic but does not connect the result to a conclusion in context).

Relational. The student integrates multiple aspects into a coherent whole (e.g., connects the choice of procedure to the data structure, interprets the result in context, and considers assumptions).

Extended abstract. The student generalizes beyond the specific task, considers alternative approaches, or critically evaluates the adequacy of the procedure.

The rubric is applied to the student's contributions only, not to the assistant's responses. Transcripts are sampled to include interactions from both courses and from multiple assignment types. The full rubric with decision rules and anchor examples is provided in the supplementary repository.

6.6 Limitations of the Study Design

Several design limitations should be noted. The sample sizes are small (total cohort $n = 14$ in Course A and $n = 18$ in Course B; T1 respondents $n = 14$ and $n = 17$; paired T1 $\rightarrow$ T3 subsamples $n = 13$ and $n = 16$), constraining statistical power and the precision of estimated trends. The study was conducted at a single institution with a single instructor, limiting generalizability. The survey instrument was developed for this study and has not been independently validated. The absence of a control group means that observed changes in attitudes cannot be causally attributed to the SMARTS intervention.

The survey was administered through a Canvas workflow that assigned each respondent an opaque course-internal identifier (distinct from institutional identifiers and not mappable to student records without access to the instructor's roster). This design preserves the privacy of individual students against incidental disclosure while permitting paired matching of responses across the three survey waves within this study. We therefore exploit the repeated-measures structure of the data directly: Wilcoxon signed-rank tests with paired rank-biserial effect sizes are applied to T1 $\rightarrow$ T3 trajectories, and Mann–Whitney U tests on difference scores are used for subgroup trajectory comparisons (Section 6.5).

Three residual limitations of the quantitative analysis warrant explicit mention. First, the paired samples are small ($n = 13$ and $n = 16$), which yields wide bootstrap confidence intervals on effect sizes; readers should weight direction and magnitude of r_{rb} point estimates against the considerable uncertainty reflected in the intervals. Second, the pre-specified subgroup analyses (Section 7.4) operate on subgroup cells as small as $n = 2$ to 5; per Mangiafico (2016), bootstrap intervals at extreme r_{bs} values or with small cell counts may be unreliable, and we flag specific comparisons where this caveat applies. Third, the subgroup comparisons are hypothesis-generating rather than hypothesis-confirming despite being pre-specified, because the pilot was not powered to detect subgroup-by-condition interactions at the effect sizes likely to exist. Null subgroup results should not be read as evidence of equivalence; wider replication is required.

Power, ceiling, and Type-M error. Two closely related features of the data constrain the inferences we can draw. First, power for detecting moderate paired effects is limited at the sample sizes available: for a matched-pairs effect of magnitude $|r_{rb}| \approx 0.30$ at $n \in \{13, 16\}$ with $\alpha = 0.05$ two-sided, approximate power under the exact Wilcoxon signed-rank test is substantially below the conventional 0.80 threshold. Non- rejection of a null is therefore consistent with either the absence of a meaningful effect or with insufficient power to detect one; we interpret borderline results accordingly and weight the compatibility intervals on effect sizes above individual p-value cutoffs. Second, because statistically-detectable effects in small samples tend to be the ones whose random sampling error happens to inflate their magnitude, small-sample studies are known to produce point estimates that exaggerate the underlying effect size (Type-M error; Gelman and Carlin, 2014). We report every tested item, bold only the three items whose bootstrap confidence intervals exclude zero, and refrain from extrapolating the point estimates as expected future effect sizes. Ceiling effects compound this constraint: at T1, many items already exceed 4.0 on the five-point scale, leaving little room for further upward movement. Notably, the item with the largest positive paired shift in Course A (Q8, feel prepared before lecture) is the item with the *lowest* T1 baseline (3.38), which is consistent with a ceiling-limited environment rather than a purely random distribution of effects.

The transcript audit relies on purposive sampling by the instructor rather than coding

the full corpus, and applies a rubric developed for this study that has not been independently validated. We release the full rubric with anchor examples in the supplementary repository to support independent verification and replication.

Researcher reflexivity. The qualitative analyses in this paper (the thematic analysis of open-ended survey responses and the SOLO transcript audit) were conducted by the instructor of both courses, who is also the designer of the SMARTS protocol and the FRAME checklist. This position brings direct course context and deep familiarity with the data, but it also introduces a risk of confirmatory bias: an investigator with prior commitment to the framework may be predisposed to identify patterns that support it. We mitigate this risk by releasing the full de-identified essay corpus, the complete thematic codebook with code frequencies, and representative student–LLM transcripts in the supplementary repository (Section 8.7), so that readers and future investigators can independently verify the coding decisions and apply alternative analytic lenses to the same data. Inter-rater reliability checks on both the thematic codebook and the SOLO rubric are deferred to future multi-coder studies.

These limitations are common in classroom-based education research and are addressed further in Section 8. The study's contribution is descriptive and generative: it documents what structured LLM use looks like in two statistics courses and provides a framework that others can adopt and evaluate at scale.

7 Results

7.1 Participants and Demographics

A total of 32 unique students consented to participate and responded to at least one survey wave, including 14 enrolled in Course A (Introductory Statistics) and 18 enrolled in Course B (Advanced Mathematical Statistics). Figure 5 presents the participation flow across the three survey waves (T1 at week 4, T2 at week 9, T3 at week 14). Twenty-five of the 32 students (78.1%) completed all three waves; the remaining attrition is consistent with ordinary class absenteeism with no wave experiencing systematic nonresponse. The strict complete-case subsample ($n = 11$ in Course A and $n = 14$ in Course B) is used for

three-wave trend sensitivity analyses reported in Section 6.5; the primary paired T1 → T3 analyses use the $n = 13$ (Course A) and $n = 16$ (Course B) students with matched endpoint observations.

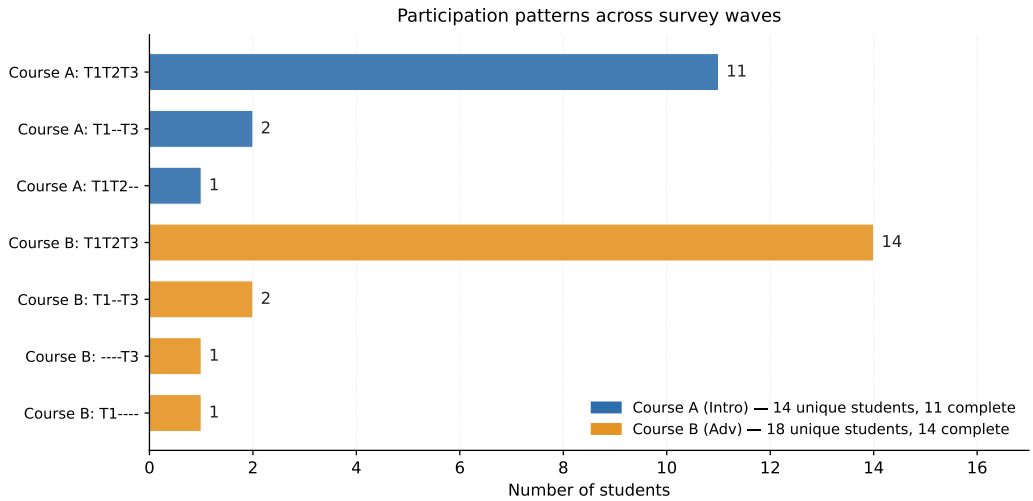

Figure 5: Participation flow across survey waves by course. Bars show the number of students following each observed attendance pattern (`T1T2T3` = all three waves; `T1--T3` = T1 and T3 only, missing T2; etc.). Course A: 14 unique students, 11 complete three-wave cases. Course B: 18 unique students, 14 complete three-wave cases. Grand total: 32 students, 25 complete cases (78.1%). Item-level non-response conditional on attendance is zero: students who took a wave answered all eleven Likert items.

Table 1 summarizes demographic survey responses (Course A: $n = 14$ responding; Course B: $n = 18$ responding) across the key demographic and technology-use questions. For items permitting multiple selections (primary language, devices, AI platforms) response counts may sum above the per-course respondent total; a small number of students did not report all single-select fields.

Table 1: Top three responses per question by course.

Question	**Response**	**Course B** n	**Course A** n
Gender	Male	11	5
	Female	7	9
Hispanic/Latino(a)	Yes	12	9
	No	6	5
Race/Ethnicity	Other	8	4
	White	3	6
	Asian	4	2
Primary language	English	14	11
	Spanish	5	3
Academic standing	Senior or later	13	1
	Junior	4	1
	Sophomore	0	5
	Freshman	0	7
Reliable computer & internet	Yes	18	14
Common devices	Laptop	15	12
	Tablet	13	3
	Smartphone	13	3
Use AI tools	Yes	17	9
	No	1	5
AI platforms used most	ChatGPT	16	9
	DeepSeek	4	0
	Google Gemini	2	2

Nearly all surveyed students reported prior independent use of LLMs for coursework or study support. ChatGPT was the most widely used platform across both courses, with particularly high adoption in Course B. A majority of participants identified as Hispanic or Latino/a (65.6%), and roughly one quarter reported Spanish as their primary language,

reflecting the institution's Hispanic-Serving designation. All students reported reliable access to a computer and the internet, and all Course B students reported prior AI tool use. In Course A, five of 14 students (35.7%) reported not using AI tools prior to the study, suggesting that the course may have served as their first structured exposure.

7.2 How Students Report Using AI and Non-AI Tools (Baseline)

Two open-ended baseline items asked students to describe how they use (i) AI tools and (ii) non-AI tools. In Course A, 12 students reported using AI tools primarily for obtaining explanations and for coding or debugging in R. Among the nine who reported using non-AI resources, the most common activities were watching video explanations and reviewing worked examples. In Course B, students reported a broader range of AI use ($n = 22$ responses), most frequently for breaking down complex problems, checking derivations, and coding or debugging tasks. Non-AI resource use in this group ($n = 20$) was dominated by consulting textbooks or online documentation and revisiting core concepts for review.

Representative excerpts:

> "AI tools help me since with my work-life-class balance they are accessible any time and can break a concept into steps." (Course B, baseline)

> "I found the R tutor helpful for explanations and code, examples and syntax explanations worked." (Course A, baseline)

7.3 Attitudes Across the Semester (Likert Items)

Tables 2 and 3 summarize paired T1 $\rightarrow$ T3 item-level results for Course B and Course A respectively, reporting wave means and standard deviations for paired respondents, the mean paired difference $\bar{\Delta}$, the tripartite count of students whose T3 response exceeded, matched, or fell below their T1 response, the Wilcoxon signed-rank statistic V with exact p-value, and the matched-pairs rank-biserial effect size r_{rb} with 95% bootstrap confidence interval. Results are reported for all 11 instrument items in both courses. Figure 6 below displays the underlying per-student trajectories visually. Full item-level data and reproduction code are available in the supplementary repository (Section 8.7).

Small-sample framing. With paired $n_A = 13$ and $n_B = 16$, the study is powered for descriptive pattern detection, not hypothesis testing. We interpret direction and effect-size magnitude with bootstrap compatibility intervals rather than p-value thresholds (Wilkinson and APA Task Force on Statistical Inference, 1999; Wasserstein et al., 2019; Bellera et al., 2010), and adopt an effect-size-first decision rule: an item is treated as having a directionally established effect when its 95% CI on r_{rb} excludes zero. This rule is more conservative than conventional null-hypothesis testing at $\alpha = .05$ for small samples, where the exact Wilcoxon test is often insensitive even to substantial paired shifts because of ties and small n. Where we report the proportion of respondents agreeing with each item, Wald-type intervals on binomial proportions are known to perform poorly near 0 or 1 at $n \approx 15$ (Agresti and Coull, 1998); Wilson-score intervals are therefore used for any %Agree summary statistics reported in the supplementary tables.

Table 2: Course B (Advanced Mathematical Statistics) student attitudes, paired T1 → T3 comparisons ($n = 16$ paired). Wilcoxon signed-rank test (exact, Pratt zero handling) with matched-pairs rank-biserial correlation r_{rb} as effect size.

Item	T1 M (SD)	T3 M (SD)	$\bar{\Delta}$	$+/0/-$	V	p (exact)	r_{rb} (95% CI)
AI tutor enhances learning	3.88 (0.62)	4.00 (0.52)	+0.12	6/7/3	39.0	0.980	+0.22 [−0.31, +0.73]
Encourages critical thinking	3.62 (0.50)	3.69 (0.60)	+0.06	4/9/3	39.0	0.433	+0.10 [−0.40, +0.55]
Helps recall/apply prior concepts	4.19 (0.75)	4.12 (0.72)	−0.06	2/11/3	28.0	0.039	−0.10 [−0.51, +0.33]
User-friendly and accessible	4.19 (0.75)	4.38 (0.62)	+0.19	6/7/3	36.0	0.860	+0.26 [−0.24, +0.73]
Feedback as helpful as instructor	3.62 (0.81)	3.50 (0.97)	−0.12	3/8/5	38.5	0.144	−0.17 [−0.64, +0.33]
Instructor prompts help	3.94 (0.77)	4.00 (0.73)	+0.06	7/4/5	56.0	0.940	+0.10 [−0.43, +0.62]
Supports learning regardless of background	4.00 (0.97)	4.12 (0.62)	+0.12	4/9/3	37.5	0.495	+0.12 [−0.37, +0.60]
Feel prepared before lecture	3.25 (0.93)	3.69 (0.87)	+0.44	6/5/5	45.0	0.706	+0.23 [−0.34, +0.69]
Understand lecture examples	3.81 (0.91)	3.75 (0.77)	−0.06	5/6/5	55.0	0.528	−0.04 [−0.55, +0.48]
Reduce external resource use	3.69 (0.79)	3.50 (0.63)	−0.19	3/7/6	39.0	0.144	−0.22 [−0.71, +0.30]
Aware of AI integration	4.38 (0.72)	4.44 (0.51)	+0.06	3/12/1	16.0	0.193	+0.19 [−0.23, +0.60]

Note. Response scale 1–5 (1 = Strongly Disagree, 5 = Strongly Agree). $\bar{\Delta}$ is the mean paired difference $T3 - T1$. $+/0/-$ counts the number of students whose T3 score exceeded, matched, or fell below their T1 score. V is the Wilcoxon signed-rank test statistic (sum of positive signed ranks) using Pratt's method for zero differences. p values are exact. r_{rb} is the matched-pairs rank-biserial correlation (Kerby, 2014); 95% confidence intervals are percentile bootstrap with 5,000 replications. Bootstrap intervals at extreme values or with small cell counts may be unreliable (Mangiafico, 2016).

Table 3: Course A (Introductory Statistics) student attitudes, paired T1 → T3 comparisons ($n = 13$ paired). Wilcoxon signed-rank test (exact, Pratt zero handling) with matched-pairs rank-biserial correlation r_{rb} as effect size. Items with r_{rb} confidence intervals excluding zero are shown in bold.

Item	T1 M (SD)	T3 M (SD)	$\bar{\Delta}$	+/0/−	V	p (exact)	r_{rb} (95% CI)
AI tutor enhances learning	4.00 (0.58)	4.08 (0.76)	+0.08	5/5/3	31.0	1.000	+0.15 [−0.41, +0.69]
Encourages critical thinking	3.54 (0.78)	3.77 (0.83)	+0.23	6/5/2	22.0	0.588	+0.35 [−0.21, +0.89]
Helps recall/apply prior concepts	3.92 (0.49)	4.31 (0.63)	**+0.38**	**4/9/0**	0.0	1.000	**+0.51** [+0.14, +0.77]
User-friendly and accessible	4.00 (0.71)	4.31 (0.63)	+0.31	3/10/0	0.0	0.542	+0.40 [+0.00, +0.69]
Feedback as helpful as instructor	3.69 (0.95)	4.08 (0.86)	+0.38	5/7/1	10.0	0.635	+0.47 [+0.00, +0.84]
Instructor prompts help	3.85 (0.80)	4.23 (0.60)	+0.38	5/7/1	10.0	0.635	+0.47 [+0.00, +0.84]
Supports learning regardless of background	4.08 (0.49)	3.77 (0.93)	−0.31	2/7/4	19.0	0.068	−0.27 [−0.71, +0.27]
Feel prepared before lecture	3.38 (0.87)	4.08 (0.76)	**+0.69**	**8/4/1**	8.5	0.068	**+0.70** [+0.24, +0.97]
Understand lecture examples	3.69 (0.75)	4.23 (0.60)	**+0.54**	**5/8/0**	0.0	0.542	**+0.60** [+0.27, +0.89]
Reduce external resource use	3.85 (0.69)	4.15 (0.55)	+0.31	5/7/1	10.5	0.685	+0.46 [+0.00, +0.84]
Aware of AI integration	3.77 (0.60)	4.00 (0.41)	+0.23	3/10/0	0.0	0.542	+0.40 [+0.00, +0.69]

Note. Columns as in Table 2. Three items (Recall/apply, Prep for lecture, Understand examples) have r_{rb} 95% CIs excluding zero, indicating directionally clear positive effects. The Equity/background item (row 7) shows a non-trivial negative shift ($\bar{\Delta} = -0.31$; 4 of 13 students rated lower at T3) with borderline p, warranting the separate subgroup analysis in Section 7.4.

Paired trajectories at the student level. Figure 6 displays the paired T1 → T3 trajectories for every student on every item, faceted by course. Green arrows indicate improvement, red arrows indicate decline, and gray dots indicate unchanged responses. The figure surfaces the paired structure visually and makes the Course A vs. Course B contrast immediately apparent: Course A's panel is dominated by rightward (green) arrows, particularly on items Q3, Q8, and Q9; Course B's panel is mixed, with comparable numbers of green and red arrows on most items.

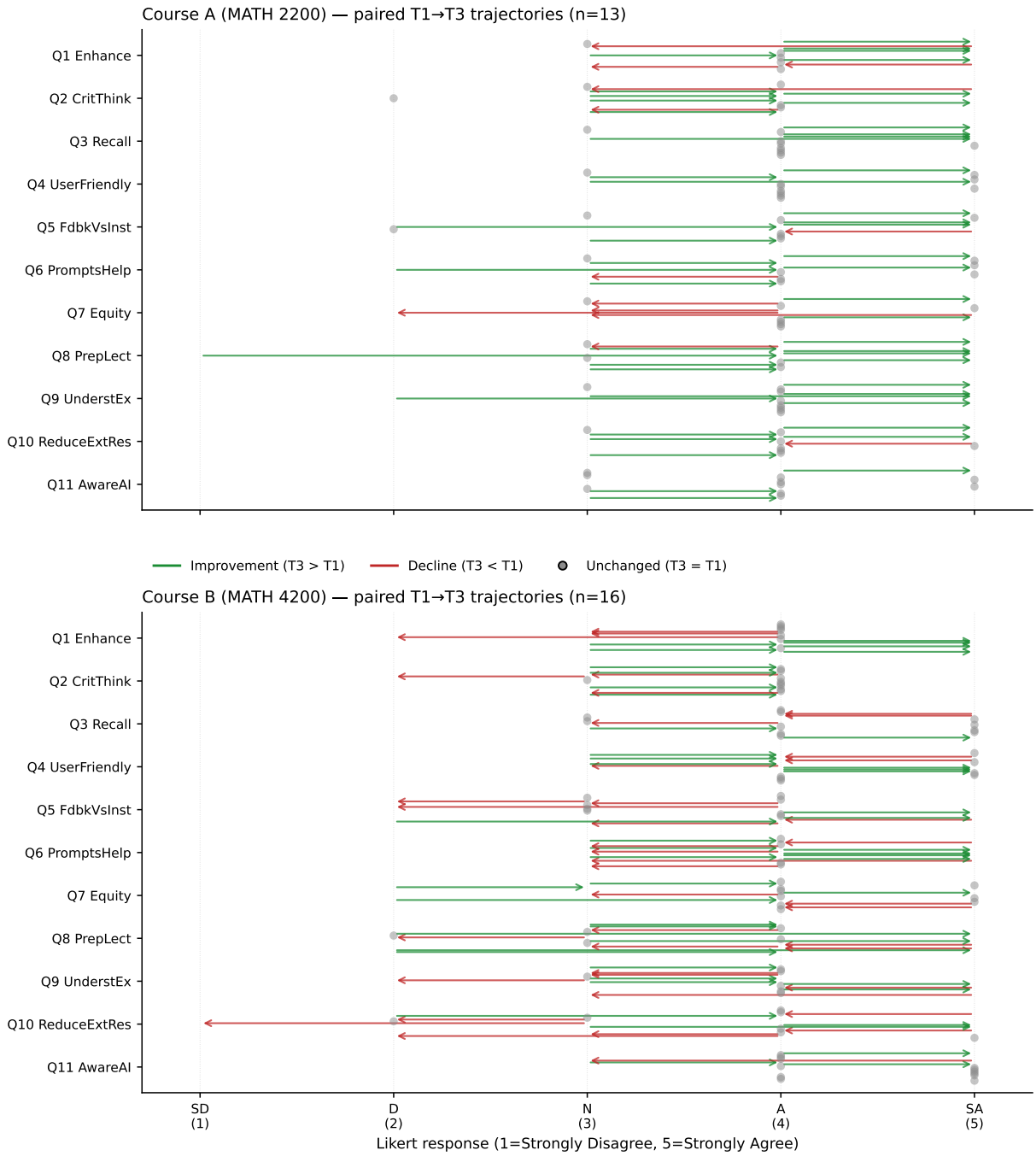

Figure 6: Individual student T1 → T3 trajectories on all 11 Likert items, by course. Each arrow is one paired student (Course A: $n = 13$; Course B: $n = 16$). Green arrows indicate an increase in agreement from T1 to T3; red arrows indicate a decrease; gray dots indicate unchanged responses. Course A's predominantly green panel on Q3 (recall/apply), Q4 (user-friendly), Q8 (prep for lecture), Q9 (understand examples), Q10 (reduce external resources), and Q11 (aware of AI integration) contrasts with Course A's red-dominant pattern on Q7 (equity/background), where 4 of 13 students rated lower at T3. Course B's mixed color pattern across all items reflects the weaker and more heterogeneous effects in the advanced course.

Course A: three items with effect-size CIs excluding zero. The strongest paired signal in the study is for *"I feel prepared before attending lecture, especially when I use prompts or AI tools to guide my preparation"* (Q8) in Course A: 8 of 13 paired students rated the item higher at T3 than at T1, 1 rated lower, and 4 were unchanged, yielding $\bar{\Delta} = +0.69$, $r_{rb} = +0.70$ (95% CI $[+0.24, +0.97]$), Wilcoxon $V = 8.5$, exact $p = .068$. The effect-size confidence interval excludes zero, making the direction unambiguous by the decision rule committed to in Section 6.5. Two additional items in Course A show the same pattern of CIs excluding zero: *"helps recall and apply previous concepts"* (Q3; $\bar{\Delta} = +0.38$, 4 improved / 9 unchanged / 0 declined, $r_{rb} = +0.51$, CI $[+0.14, +0.77]$) and *"can understand examples covered in lecture"* (Q9; $\bar{\Delta} = +0.54$, 5 / 8 / 0, $r_{rb} = +0.60$, CI $[+0.27, +0.89]$). All three items are directly connected to pre-class preparation and in-lecture comprehension—exactly the pedagogical loop the SMARTS protocol is designed to support. The remaining eight items in Course A show positive point estimates in seven cases and a negative estimate on Q7; in all eight, the 95% CI for r_{rb} crosses zero and we do not interpret the direction as established.

Course A: the equity item declined. The exception to Course A's generally positive pattern is the item *"The AI tutor supports my learning regardless of my background or prior experience with similar tools"* (Q7). Mean response decreased from 4.08 at T1 to 3.77 at T3 ($\bar{\Delta} = -0.31$; 2 students rated higher, 7 unchanged, 4 rated lower at T3). The paired Wilcoxon test returned $V = 19.0$ with exact $p = .068$, and the rank-biserial effect size was $r_{rb} = -0.27$ with 95% CI $[-0.71, +0.27]$. The confidence interval spans zero, so by our decision rule this is not a directionally established effect; however, it is also the only item in Course A for which four students lowered their ratings, and the borderline p and negative sign are not consistent with regression toward a ceiling since the T1 mean (4.08) leaves substantial room to move in either direction. We report this finding honestly and investigate it further in the subgroup analyses below and in cross-reference with the thematic analysis (Section 7.5, Theme A3).

Course B: weaker and more variable paired effects. Course B's paired results are systematically more modest than Course A's. No item in Course B has an r_{rb} 95% CI

excluding zero, so no item in the advanced course meets the effect-size decision rule. The largest mean shift in Course B is for Q8 (prep for lecture) at $\bar{\Delta} = +0.44$ (6 improved, 5 unchanged, 5 declined), echoing Course A's Q8 pattern directionally but with more students showing decline and a much wider effect-size interval ($r_{rb} = +0.23$, CI $[-0.34, +0.69]$). Q10 (reduce external resources) and Q5 (feedback as helpful as instructor) show modest negative trends ($\bar{\Delta} = -0.19$ and -0.12 respectively), neither with CI excluding zero. Q3 (recall/apply prior concepts) returned a nominally significant exact $p = .039$ with $V = 28.0$; this result is driven by the signed-rank structure of 11 tied pairs among 16 paired students and corresponds to $\bar{\Delta} = -0.06$ with $r_{rb} = -0.10$ (CI $[-0.51, +0.33]$), so the reported p should be read as a small-sample tied-data artifact rather than evidence of an effect. The substantively honest summary of Course B is that we do not have the evidence to claim paired attitudinal change in either direction on any item in this course.

The course-level contrast as a finding. The contrast between Course A's three CI-excluding-zero effects and Course B's zero is itself a finding. Course A is an introductory statistics course in which the AI assistant is most directly useful as a scaffold for the interpretive and conceptual work that the course is designed to teach. Course B is an advanced mathematical statistics course in which the assistant is operating under an unusually restrictive content rule (no proofs, no derivations, no complete solutions) against a curriculum where proof construction is the principal activity. The weaker paired signal in Course B is consistent with the hypothesis that the scaffolded-LLM approach has more pedagogical traction at lower levels of statistical content—where the protocol's gentle redirection toward student reasoning matches the form of the course's intended thinking—and less traction in proof-heavy courses, where the content rule removes most of what students can productively ask for. This is a hypothesis the pilot generates, not confirms; wider replication across course levels is required.

7.4 Subgroup Trajectories

Four pre-specified subgroup analyses were conducted per course, each testing whether the T1 $\rightarrow$ T3 trajectory on a theoretically motivated item or composite differs between two

demographic subgroups (Section 6.5). Each comparison uses the exact Mann–Whitney U test on the T3 − T1 difference scores between subgroups, with Glass rank-biserial r_{bs} as the effect size and a 5,000-replication percentile-bootstrap 95% CI. Table 4 summarizes all eight analyses.

Table 4: Pre-specified subgroup analyses of paired T1 → T3 difference scores. Each row reports a Mann–Whitney comparison of difference scores between two demographic subgroups within one course. $\bar{\Delta}$ is the mean paired change within each subgroup, and $+/0/-$ is the tripartite count. r_{bs} is the Glass rank-biserial effect size; CIs are 5,000-replication percentile bootstrap.

Analysis	Group 1	Group 2	U	p	r_{bs} (95% CI)
A1: Hispanic/Latino identity × Q7 (equity/background)					
Course A	Hispanic/Latino n=8, $\bar{\Delta}$= − 0.12, 1/6/1	Non-Hispanic n=5, $\bar{\Delta}$= − 0.60, 1/1/3	26.0	0.435	+0.30 [−0.40, +1.00]
Course B	Hispanic/Latino n=10, $\bar{\Delta}$= + 0.30, 3/6/1	Non-Hispanic n=5, $\bar{\Delta}$= + 0.00, 1/3/1	29.5	0.679	+0.18 [−0.38, +0.66]
A2: Prior AI tool use × composite Q1–Q11 mean					
Course A	Prior AI user n=8, $\bar{\Delta}$= + 0.28, 4/3/1	No prior AI n=5, $\bar{\Delta}$= + 0.31, 3/0/2	17.0	0.724	−0.15 [−1.00, +0.65]
Course B	Prior AI user n=14, $\bar{\Delta}$= + 0.03, 8/1/5	No prior AI n=1, $\bar{\Delta}$= + 0.36, 1/0/0	3.5	0.667	−0.50 [−0.86, −0.07]†
A3: English as primary language × Q2 (critical thinking)					
Course A	English primary n=9, $\bar{\Delta}$= + 0.11, 4/3/2	Other n=4, $\bar{\Delta}$= + 0.50, 2/2/0	15.0	0.710	−0.17 [−0.67, +0.44]
Course B	English primary n=10, $\bar{\Delta}$= + 0.10, 2/7/1	Other n=5, $\bar{\Delta}$= + 0.20, 2/2/1	22.5	0.859	−0.10 [−0.72, +0.52]
A4: Employment hours × Q8 (prep for lecture)					
Course A	Not employed n=3, $\bar{\Delta}$= + 1.67, 3/0/0	Working 20+ hrs n=2, $\bar{\Delta}$= + 0.00, 0/2/0	6.0	0.200	+1.00 [+1.00, +1.00]†
Course B	Working 20+ hrs n=5, $\bar{\Delta}$= − 0.60, 0/2/3	Working <20 hrs n=3, $\bar{\Delta}$= + 0.00, 1/1/1	4.5	0.571	−0.40 [−1.00, +0.40]†

Note. †Cells or effect-size estimates flagged as potentially unreliable per Mangiafico (2016): (a) Course B A2 contains only $n = 1$ in the no-prior-AI group; (b) Course A A4 $r_{bs} = +1.00$ indicates complete rank dominance (the two groups' difference-score ranges do not overlap), which yields a degenerate bootstrap interval; (c) Course B A4 has a small cell ($n = 3$) in the Working <20 hrs group. We report all analyses for completeness but caution against strong inference from these three comparisons in particular.

A1: Equity by ethnicity diverges between courses. The only subgroup effect with $|r_{bs}| \geq 0.3$ in either course was the equity item (Q7) stratified by Hispanic/Latino identity in Course A, where non-Hispanic students (3 of 5 declined) drove the overall Q7 decline observed in Course A, while Hispanic/Latino students remained approximately stable (only 1 of 8 declined). The same comparison in Course B runs in the opposite direction: Hispanic/Latino students showed positive mean change ($\bar{\Delta} = +0.30$) while non-Hispanic students were flat ($\bar{\Delta} = +0.00$). Neither comparison has a confidence interval excluding zero, and the cells are small, so these patterns are hypothesis-generating rather than confirmed. The course-level directional reversal on the same item is, however, striking, and it suggests that equity perception may depend on course context (course level, subject matter, student composition) in ways that a single-course or single-demographic analysis cannot reveal. We return to this observation in Section 8.3.

A2: Prior AI use does not distinguish composite-score trajectories in Course A. The most substantively important null result in the study is for Course A composite Q1–Q11 stratified by prior AI tool use: students who reported prior AI tool use and students who did not showed essentially identical mean composite shifts (prior users: $\bar{\Delta} = +0.28$, 4/3/1; no prior AI: $\bar{\Delta} = +0.31$, 3/0/2), with $r_{bs} = -0.15$ and a CI spanning nearly the full range. This null is directly informative: the paper's central equity claim is that structured prompt engineering via FRAME and SMARTS can equalize access to AI-supported learning across students with and without prior experience with AI tools. The data are consistent with this claim in the introductory course. The Course B comparison is uninformative because only a single student in Course B reported no prior AI use (the Course B flag in Table 4); we report the number for completeness but do not interpret the Course B A2 result.

A3: Language subgroup differences are directionally suggestive but uncertain. In both courses, non-English-primary-language students showed slightly larger positive shifts on the critical-thinking item than English-primary students (Course A: $\bar{\Delta} = +0.50$ vs. $+0.11$; Course B: $+0.20$ vs. $+0.10$), but both effect sizes ($r_{bs} = -0.17$ and -0.10) are small and both CIs span zero. Replication in larger samples is warranted before attaching

substantive weight to this pattern.

A4: Employment shows course-dependent patterns on prep-for-lecture. In Course A, non-employed students drove the Q8 (prep for lecture) gain: all 3 non-employed students increased their rating from T1 to T3 (mean increase of 1.67 Likert points), while both working-20+-hr students were unchanged. This is the opposite of the naive time-poverty prediction that working students should benefit most from AI-assisted preparation. The confidence interval is degenerate ($r_{bs} = +1.00$, CI $[+1.00, +1.00]$) because the two groups' rank distributions do not overlap, and the cells are small ($n = 3, 2$); the observation is suggestive and should not be over-interpreted. In Course B, Working 20+ hr students showed a negative Q8 trajectory ($\bar{\Delta} = -0.60$, 0/2/3), again with small cells. The two course-level patterns are inconsistent with each other, and we interpret the A4 results primarily as a reminder that workload interacts with AI-supported preparation in ways the present study is not powered to disentangle.

7.5 Open-Ended Responses: Thematic Analysis

The 428 open-ended responses were coded following the hybrid inductive–deductive procedure described in Section 6.5. Separate thematic analyses were conducted for each course. Course A yielded five themes spanning the five prompt domains; Course B yielded seven themes with two additional themes (critical engagement with AI and memory offloading) that did not appear in Course A. Table 5 summarizes the themes and approximate code-instance totals; the full codebook, all coded extracts, and per-wave code frequencies are released in the supplementary repository (Section 8.7).

Table 5: Themes from the thematic analysis of open-ended responses, by course. Code instances are approximate counts of text segments tagged with codes belonging to each theme; a single response may contribute to multiple themes through multi-coding.

ID	Theme	Instances
Course A (Introductory Statistics)		
A1	Time-efficient, on-demand scaffolding	57
A2	R code assistance as distinctive course-level affordance	22
A3	Equity: unconditional at baseline, prompting-conditional by end	41
A4	Critical-thinking trade-off: helps and hinders	13
A5	Suggestions concentrated on constraint, not freedom	36
Course B (Advanced Mathematical Statistics)		
B1	On-demand theorem and definition retrieval	13
B2	Course-aligned instruction as the critical affordance	13
B3	Hint-privileging interaction style as central design concern	25
B4	Critical engagement with AI as cognitive work	9
B5	Memory and cognitive offloading ambivalence	11
B6	Equity as effectively universal, qualifications low-salience	46
B7	Sophisticated suggestion-making at the pedagogical-design level	15

Note. Theme totals reflect code instances, not unique respondents; the denominator for per-theme prevalence varies by prompt. Codebook in the supplementary repository documents the code-to-theme mapping in full.

Course A themes

A1: Time-efficient, on-demand scaffolding. The dominant benefit frame across all five prompt domains described the assistant as a 24/7 available scaffolding resource that compresses time-to-understanding. Concrete instances included catching up after missed class sessions (for sport travel, illness), step-by-step debugging of R code, and generating mock exam questions for review. The theme was stable across all three waves.

> "AI tools help my ability to manage and succeed by helping me remember concepts that I might have forgotten. They help refresh my brain." (T1_2200_14)

> "I could ask it to provide me with practice questions and thats the way I practiced for the final exam." (T3_2200_06)

> "AI tools help me by remembering certain steps from the lesson that I missed. Or if I missed a class due to sports traveling it helped me catch up too." (T3_2200_03)

A2: R code assistance as the distinctive course-level affordance. The most frequently praised specific feature in Course A was R coding help. References to "the R tutor" became more explicit and more frequent at T3, suggesting consolidation of the tool's identity across the semester. Several responses contrasted the course-aligned R tutor with generic AI, praising the former's restriction to the `mosaic` package convention used in the course.

> "i really like that R-Studio feature the most help full, at first i didn't know how to deal with R but by the end of the course AI tutor has taught me they all that codes the professor showed that i forgot." (T3_2200_01)

> "The R tutor explanations are effective, but some in-class examples required much simpler versions of the code, thus using the custom GPT helped to narrow down the codes." (T3_2200_09)

A3: Equity as near-universal, prompting-conditional at end of semester. At baseline, Course A students overwhelmingly affirmed unconditional equity: "I think that the AI tutor accommodates the needs of all the students equally. I don't think there is any bias" (T1_2200_14). By the end of the semester, a subset of students articulated a more

skill-mediated view: realized equity depends on students' prompting skill, which in turn depends on the instructor's explicit teaching of prompt engineering.

> "I do not think AI accommodates the needs of all students equally; I think a student's effectiveness at utilizing prompts determines the quality of a student's experience with AI." (T3_2200_09)

> "Yes, if you are taught how to use/prompt the tool." (T3_2200_10)

This longitudinal shift is modest in absolute frequency but substantively meaningful: students who have spent a semester using the tool articulate a skill-mediated rather than access-mediated view of equity, which is consistent with the paper's argument that prompt-engineering discipline (operationalized in this study as FRAME) is itself an equity lever.

A4: Critical-thinking trade-off. A subset of students articulated the assistant's impact on critical thinking as a structural trade-off: the same scaffolding that helps at the point of confusion can, if used indiscriminately, short-circuit the effortful struggle that constitutes learning. Responses at T1 already contained this awareness in some students; responses at T3 made the concern more specific.

> "AI ends up being a shortcut to learning the material, which I think ends up hurting the student more than helping them actually understand the material." (T1_2200_13)

> "At times I would find myself leaning on AI too much; if I needed to solve a problem for a test, I would find myself lacking the critical thinking needed to tackle certain problems." (T3_2200_09)

> "It makes it harder to want to actually learn the material when you know you can just search up the answer." (T3_2200_10)

A5: Suggestions concentrated on constraint, not freedom. Of 37 substantive responses to the suggestions prompt across waves, 29 were "none / works great." Among students who did offer substantive suggestions, the common thread was a desire for more constraint (keep output relevant to the course, limit scope, improve accuracy, standardize deployment across classes) rather than less. Only one student at T1 requested fewer restrictions.

Course B themes

B1: On-demand theorem and definition retrieval. A course-distinctive use case absent from Course A emerged: students described using the assistant as a rapid theorem-and-definition retrieval system during problem solving, substituting AI-mediated lookup for textbook page-flipping.

> "AI tools such as ChatGPT reduce time for me flipping through my textbook looking for the certain theorems and definition that I need to solve a problem." (T3_4200_13)

> "It helps recall a lot of things like definition, formulas and even symbols especially when it came to not being time consuming looking through the notes." (T3_4200_11)

B2: Course-aligned instruction as the critical affordance. Course B students repeatedly identified course-alignment—the assistant's use of the professor's notation, the textbook's definitions, and the lecture's worked-example conventions—as the feature that made the tool pedagogically useful rather than confusing. The theme was expressed most clearly in contrast.

> "I didn't like using AI before because it seemed like it would always try to teach me a different way that I wasn't taught in class and I would end up more confused." (T1_4200_11)

> "The most helpful thing about the AI tutor was that the teaching style and notation was exactly like the professor's." (T3_4200_12)

B3: Hint-privileging interaction style as central design concern. The most frequently articulated improvement request across both the features prompt (Q13) and the suggestions prompt (Q16) concerned the assistant's tendency to give full answers when students wanted hints. When the hint-privileging mechanism worked, students explicitly valued it; when it failed, students criticized it. Several students observed the instructor iterating on the system prompt across the semester to reduce answer-giving, indicating that the refusal mechanism functions as a visible pedagogical contract rather than invisible infrastructure.

> "At the beginning it would just give the answers but once the professor fixed it has helped me think more about the problems." (T2_4200_09)

> "I found the AI's abilities to only provide small hints and partial steps really useful in helping me learn." (T3_4200_07)

> "I find it most help that I could ask the AI tutor any question that I might feel too embarrassed to ask my actual professor. I also like that I can 'think out loud' to the AI tutor and it will correct where my thought process might have gone wrong. I think a feature that needs improvement is not giving into the answer too quickly because you can ask it to not give you the answer, but after about two more chats expressing confusion, the AI tutor gives the full answer." (T3_4200_09)

B4: Critical engagement with AI as cognitive work. A distinctive pattern appeared in Course B that was not detected in Course A: students described arguing with, questioning, or critically evaluating the AI's output as a cognitive exercise in its own right. Unlike Theme A4's worry that the tool displaces thinking, Theme B4 captures students using the tool to sharpen thinking by contesting its claims.

> "It has really helped me think critically. There have been many times where I have questioned its results and had to argue with it." (T3_4200_12)

> "I came across a question with the last version that the bot could answer because it had the solution but it could not explain the solution to me. After arguing with it, I gave up and asked Dr. Y and he was able to explain." (T2_4200_10)

The theme is rare in the corpus—approximately five to six students—but consistent in pattern and present at all three waves. It is pedagogically notable because the posture (student as critical evaluator of AI output) emerges without explicit scaffolding, driven by the combination of advanced subject matter and a course-aligned but fallible tool.

B5: Memory and cognitive offloading ambivalence. By the end of the semester, a cluster of Course B students articulated that AI use had materially weakened their ability to recall course material without the assistant. Unlike Theme A4's general concern about over-reliance, these students named specific losses: memory for formulas, confidence in

solving without reassurance, and what one student called "offloading my memory to the AI."

> "I have a very busy life so using the AI does help me get through the lessons ... However, as it is helping me recall what I would need to know in order to complete the assignment, I do feel like my memory retention for courses has been impacted greatly ... I am unable to recall formulas and theorem with ease." (T1_4200_10)

> "I do feel that overtime it definitely has hindered my ability to think critically because there is an ai tutor crutch that I can use whenever I get stuck. Instead of genuinely thinking of how to get unstuck on a problem and therefore learning in the process I have a tendency to use the tutor and skipping that critical thinking completely." (T3_4200_07)

> "Sometimes feel like im offloading my memory to the AI and rely on it. This hinders my learning because im overly relying on it for guidance and too lazy to put actual practice and test my mind." (T3_4200_13)

The theme is notably absent at T1 and appears most strongly at T3, paralleling external empirical findings on cognitive offloading (Gerlich, 2025) and reduced critical thinking associated with confidence in generative AI (Lee et al., 2025).

B6: Equity as effectively universal with low-salience qualifications. Course B students overwhelmingly viewed the assistant as equitable. Qualifications existed but were less prominent than in Course A. Language support appeared as a named benefit rather than a gap for several students.

> "Yes, since it cannot treat you differently and can even communicate in different languages." (T3_4200_01)

> "Yes, because I usually struggle with English and AI has helped understand better what we are doing in class." (T3_4200_08)

Only a minority of Course B responses expressed the prompting-skill concern that emerged more prominently in Course A. This pattern is consistent with the broader literature (Rao et al., 2026; Krupp et al., 2024) that unstructured AI use benefits students

with stronger foundations disproportionately: Course B students are mostly mathematics majors with stronger prior preparation and consequently articulate equity concerns less sharply.

B7: Sophisticated suggestion-making at the pedagogical-design level. While the majority of Course B suggestion responses were "none," the substantive suggestions that appeared were notably more pedagogically sophisticated than in Course A. Students proposed specific assignment-design modifications (frequency, size, format), integration with course artifacts (class notes, PDFs, textbook), and design-level features (Socratic opener, visual support).

> "It would be helpful if there were more assignments with tutor assistance but less questions on each assignment. Ex: Having 10 AI assisted assignments throughout the course 1 every friday ... with about 3–4 questions each assignment in order for students to keep that familiarity with using the tool and so that students can have weekly practice with problems as well instead of having only a few large question assignments." (T3_4200_01)

> "Id say that one of the best things that could be done are to have a step by step solution for every problem if possible so that the AI tutor wont contradict itself and perhaps ask the students a few questions about what they do know about the subject if they are just completely lost on how to start a problem instead of just showing the whole answer." (T2_4200_07)

Cross-course convergences and divergences

Four patterns recurred across courses: on-demand scaffolding (A1/B1+B2), hint-privileging concerns (more articulated in B3 but present in A), critical-thinking trade-off (A4/B5), and equity as near-universal (A3/B6). Three patterns were course-distinctive: R code help as the dominant feature in Course A (Theme A2), theorem-and-definition retrieval as the dominant feature in Course B (Theme B1), and critical engagement with AI as cognitive work (Theme B4) present only in Course B. We return to the pedagogical implications of these divergences in Section 8.

7.6 Transcript Audit Findings

To complement the attitudinal survey, we report an instructor-conducted audit of a purposive sample of student–LLM transcripts coded with the SOLO-adapted rubric described in Section 6.5. The aims of this audit are two: (i) to characterize the range of reasoning depths elicited in practice, and (ii) to document whether the assistants' refusal and escalation logic is observable in transcripts where a student's opening turn requested a complete solution.

Sampling and procedure. Transcripts were drawn from assignment submissions across both courses, sampled to span assignment types (R troubleshooting, hypothesis-testing lab work, and proof-based problem sets in Course B), assignment phases within the semester, and the full range of perceived engagement quality based on preliminary review. All transcripts were de-identified before coding. The instructor (author AY) applied the rubric to the student's highest-level reasoning turn in each transcript, with a one-sentence justification recorded against the anchor examples for each level. We note plainly that the rubric was applied by a single coder who also served as course instructor; inter-coder reliability was not assessed in this initial audit, and the results are therefore best read as illustrative rather than as a reliability-quantified distribution. Formal reliability work with independent coders is identified as a priority in Section 8.

Distribution of reasoning depths in the annotated examples. The four annotated examples documented in Section 4.3 (Lipitor one-sample proportion test; one-way ANOVA with Tukey HSD; unbiasedness of $\hat{\theta} = nY_{(1)}$; unbiasedness of S^2), together with the additional illustrative sessions reproduced in Appendix D, were coded using the SOLO-adapted rubric. In every annotated example, the student's final turn reached the *Relational* level: the student integrated the procedural output with the underlying inferential rationale (hypothesis-test decision rule; family-wise error rate across the six Tukey pairwise comparisons; moment-generating-function argument for the order-statistic estimator; expectation structure of the sample variance). No example in the annotated set reached the *Extended Abstract* level, which would require the student to generalize the procedure to a distinct but related problem without further prompting; this absence is consistent with

the undergraduate level of both courses and with the scaffolded rather than generative nature of the SMARTS design. We do not claim that this distribution is representative of the full transcript corpus: the annotated examples were selected to illustrate successful engagements with the protocol, and a systematic population-level distribution across all audit transcripts with independent coders would be required to characterize the full range of SOLO levels observed in practice. We identify the full systematic coding, with inter-rater reliability, as a priority for follow-up work (Section 8).

Observations on the refusal-and-escalation logic. Across the audited transcripts, a recurring pattern was visible in interactions where the student's initial turn was an answer-extraction request (for example, a pasted problem with "solve this" appended). In these cases the assistant's scaffolded response—typically a paraphrase of the task followed by a guiding question or a request for the student's attempted approach—did not always produce immediate compliance with the SMARTS phase structure, but it did consistently redirect the conversation away from a single-turn solution. In several transcripts the student re-engaged after two or three assistant-led prompts, producing written reasoning that would code at the Multistructural or Relational level under the rubric. This pattern corroborates the protocol's design intent as described in Section 4.2: the assistant is not relying on students to internalize and apply SMARTS deliberately, but rather exerts scaffolding pressure through its own turn-by-turn behavior. We view student non-adherence to the SMARTS phase sequence as expected and informative rather than as a failure of the protocol; the measurable question is whether the assistant's structure shifts the trajectory of the conversation, and the audit suggests it often does.

What the audit does not establish. Because the sample is purposive and single-coded, we do not report a population-level distribution of reasoning depths. We do not claim that all or most student–LLM interactions in these courses reached the Relational level, and we do not use the audit to infer causal effects of the protocol on reasoning quality. The audit's value is illustrative and definitional: it makes concrete what the rubric codes for, what the assistants' redirection looks like in live student dialogue, and what a larger, independently coded study could measure.

8 Discussion, Limitations, and Future Work

8.1 Summary of Findings

The results address the three research questions as follows. Regarding RQ1, student attitudes toward the SMARTS-governed assistants were positive at baseline and, in the introductory course, shifted further in the expected direction on pedagogically central items. Three Course A items had rank-biserial effect-size confidence intervals that excluded zero: feeling prepared before lecture ($r_{rb} = +0.70$, 95% CI $[+0.24, +0.97]$), understanding examples in lecture ($r_{rb} = +0.60$, $[+0.27, +0.89]$), and recalling/applying prior concepts ($r_{rb} = +0.51$, $[+0.14, +0.77]$). Course B showed directionally similar but weaker paired shifts, with no item reaching the same effect-size threshold; the course-level contrast is consistent with the hypothesis that SMARTS's traction is greater in lower-level statistical content than in proof-centered advanced courses. The equity item declined in Course A ($r_{rb} = -0.27$), with non-Hispanic students driving the overall decline in the pre-specified subgroup analysis (Section 7.4); we report this finding honestly rather than selectively and interpret it in light of the study's sample size (Section 6.6). Regarding RQ2, open-ended responses revealed convergent themes: students valued on-demand availability, step-by-step explanations, and R troubleshooting support (Course A) or definitional and conceptual scaffolding (Course B). Students also identified a tension between scaffolding benefits and the temptation to use the assistant for answer extraction—a finding that corroborates the design rationale for the refusal-and-escalation mechanisms built into SMARTS. Regarding RQ3, instructor-conducted audits of a purposive sample of student–LLM transcripts document heterogeneous reasoning depth across the SOLO hierarchy and illustrate the assistants' redirection logic in cases where students' initial turns requested complete solutions (Section 7.6). Systematic coding of the full transcript corpus with independent coders and formal inter-coder reliability is deferred to future work; the coding rubric with anchor examples is released with this paper to support that work (see Section 8.7).

8.2 Connections to Statistical Thinking and GAISE

The findings are consistent with the theoretical framing developed in Section 2. Wild and Pfannkuch's (1999) framework identifies the investigative cycle and the interrogative cycle as central to statistical thinking. The SMARTS protocol was designed to operationalize both: the State and Map phases correspond to the Problem and Plan stages of the investigative cycle, while the Think phase targets the generate–seek–interpret–criticize–judge interrogative loop. The FRAME checklist embedded in Phase 1 further reinforces the investigative cycle by requiring students to articulate facts, objectives, and assumptions before engaging the AI, mirroring the problem-formulation skills that the GAISE report identifies as foundational (GAISE College Report ASA Revision Committee, 2016).

8.3 Equity: A Mixed Finding Requiring Careful Interpretation

The observation by Rao et al. (2026) that generative AI tools primarily benefit students with strong prior understanding motivated the equity design elements of SMARTS: the explicit FRAME checklist to support students without prior prompting experience, the content-neutral cross-cutting rule to prevent the assistant from widening gaps through selective shortcut-taking, and the instructor audit loop to detect and correct drift. Our data offer a mixed rather than uniformly confirmatory picture of whether these design elements achieved their equity intent.

On the supportive side, the pre-specified subgroup analysis of composite Q1–Q11 trajectories by prior AI tool use (Section 7.4, analysis A2) returned a near-zero effect-size estimate in Course A: students with prior AI experience and students without showed essentially identical mean composite shifts (Prior AI: $\bar{\Delta} = +0.28$; No prior AI: $\bar{\Delta} = +0.31$), consistent with the claim that structured scaffolding can equalize access across students with different starting points.

Less favorably, the Q7 equity item itself— *"The AI tutor supports my learning regardless of my background or prior experience with similar tools"*—declined in Course A ($\bar{\Delta} = -0.31$; 4 of 13 students rated lower at T3), and the pre-specified Hispanic/Latino × Q7 subgroup comparison showed that non-Hispanic students drove this decline (3 of 5 non-Hispanic students declined, vs. 1 of 8 Hispanic/Latino). This pattern was *absent* in Course B, where

the Q7 item shifted mildly positive on average with Hispanic/Latino students showing a modest increase. The course-level directional reversal on a single item, pre-specified before analysis, cannot be dismissed as multiple-testing noise and also cannot be read as a confirmatory finding given the small cell sizes (Non-Hispanic $n = 5$ in Course A). It is, however, consistent with the interpretation that equity perception emerges not only from the assistant's behavior but also from each student's sense of who the protocol is designed for, and that this sense can be course-dependent. One plausible reading is that introductory-course students, who meet SMARTS without prior expectations about what an AI tutor should do, are more sensitive to the protocol's constraints than advanced-course students, who bring a firmer sense of what constitutes appropriate AI help for their course material. This reading is speculative; deliberate investigation with larger, more diverse samples is required to distinguish it from alternative explanations.

The qualitative thematic analysis provides triangulation (Section 7.5, Theme A3): Course A students at T3 referred to effective AI use as contingent on prompting skill, which some classmates did not share equally. This pattern, visible in the essay corpus, aligns directionally with the quantitative Q7 decline in Course A. The qualitative and quantitative findings converge on the same substantive concern: at the introductory level, SMARTS may help students with strong prompting skills more than those without, the opposite of the protocol's equity intent for that population. Diagnostically, this argues for more explicit FRAME instruction earlier in the semester and for periodic mid-course equity check-ins, rather than for abandoning the scaffolded approach.

8.4 Practical Implications for Statistics Instructors

The SMARTS protocol and the FRAME checklist are designed to be portable. They require no custom software, no programming, and no institutional license beyond access to a standard LLM interface. An instructor wishing to adopt the approach would need to: (a) prepare a set of course-specific materials (lecture notes, worked examples, approved problem sets) to upload as the Custom GPT's knowledge base; (b) embed the SMARTS system instructions and refusal/escalation clauses in the assistant's configuration; (c) conduct an AI literacy session introducing students to LLM capabilities, limitations, and the distinc-

tion between answer extraction and guided inquiry; (d) distribute the SMARTS/FRAME reference card; and (e) require dual submission (solution plus transcript) for AI-assisted assignments. All of these materials are provided in the supplementary repository.

Several practical considerations emerged from the deployment. First, the AI literacy session was essential. Students who understood how LLMs generate output were more receptive to the rationale for structured prompting and more willing to invest effort in FRAME-structured queries. Second, prompt maintenance is ongoing. LLM updates occasionally altered the assistant's behavior in subtle ways, requiring the instructor to review and adjust system instructions mid-semester. Third, a minority of students (approximately 22% in one assignment analysis) initially attempted to circumvent the scaffolding by restarting conversations or using unconstrained tools. Individual conversations emphasizing the learning rationale, rather than punitive measures, were effective in redirecting most of these students.

Framing the moment: abstraction, not rupture.

The anxieties surrounding LLMs in education often rest on an implicit framing of generative AI as a rupture with prior educational practice. A longer view suggests a different framing. The history of statistical computing is a history of rising abstraction: from binary machine code through punched cards, assembly, FORTRAN, C, C++, Java, and Python, to statistical languages such as R and Julia that assume the user thinks in models rather than in memory addresses. Each successive layer narrowed the gap between the analyst's statistical intent and the machine's execution. Natural-language interfaces to statistical computation, of the kind that LLMs now enable, are best understood as the next rung on this staircase rather than a break from it. A student in an introductory course can today describe a statistical task in English and receive executable R, in the same way that a student in the 1980s could describe a regression in SAS syntax and bypass the assembly-level code that would have been required a generation earlier. The pedagogical question this raises is not whether the abstraction is legitimate but what statistical competencies the layer below now requires students still to understand. Our position, embodied in SMARTS and FRAME, is that the required competencies are the ones that never sat

inside any programming language: problem formulation, assumption articulation, method selection, and interpretation in context. Those are exactly the competencies the GAISE report identified as foundational long before LLMs existed (GAISE College Report ASA Revision Committee, 2016; Wild and Pfannkuch, 1999).

Framing the moment: equilibrium, not displacement.

A parallel framing applies to labor-market concerns that students and faculty raise when AI enters the classroom. Public discussion of AI and employment tends to oscillate between displacement narratives and boosterism. Neither captures the more mundane pattern observed when the internet became pervasive in the late 1990s and early 2000s: some occupations contracted, many were reshaped, and a larger set of new roles appeared at the interface between the new technology and the domains it touched. We expect an analogous equilibrium to emerge for statistical and data-analytic work, in which routine procedural tasks are increasingly delegated to AI assistants while the demand for analysts who can *specify* problems well, audit AI outputs, and translate between stakeholders and models grows. This expectation is not a forecast and we do not defend it empirically here; it is a framing assumption that shapes the pedagogical commitments of the SMARTS protocol. If the long-run equilibrium is one in which statistical professionals spend more time on specification, auditing, and interpretation and less on syntax, then a statistics education that trains students to formulate problems precisely (FRAME), to reason stepwise through an investigative cycle (SMARTS), and to audit AI interaction quality (the transcript-review practices described in Section 5) is aligned with what the next generation of statistical work will reward, regardless of how the specific mix of tools evolves.

8.5 Limitations

This study has several limitations that constrain the strength of its claims. The sample sizes ($n = 14$ and $n = 17$) limit statistical power and the precision of estimated trends; the observed stability of attitudes may partly reflect insufficient power to detect small changes. The study was conducted at a single institution with a single instructor, and the results may not generalize to other institutional contexts, student populations, or instructors.

The survey instrument was developed for this study and has not undergone independent psychometric validation; future work should establish its reliability and factorial structure with larger samples. The absence of a control group means that attitudinal trends cannot be causally attributed to SMARTS rather than to other instructional factors such as the instructor's teaching style, the course progression, or maturation effects. The transcript audit relies on purposive sampling and instructor coding; inter-rater reliability should be established in future studies by involving independent coders.

The stochastic nature of LLM outputs is an additional concern. Even with system instructions and constrained knowledge bases, the same prompt can produce different responses on different occasions, creating variability in the student experience. We mitigated this through transcript review and iterative refinement of the system instructions, but the issue is inherent to current LLM technology and will require ongoing attention.

8.6 A Broader Portfolio of Course-Aligned Assistants

The SMARTS-governed assistants described in this study sit within a broader portfolio of course-aligned Custom GPTs that the instructor has developed and deployed in statistics and mathematics courses at the same institution. Each is built on the same underlying design commitment, namely that a narrow, deliberately constrained assistant is pedagogically more useful than a generic one, but each solves a different instructional problem. We describe the portfolio briefly here to situate SMARTS within a wider practice of instructional design rather than as a standalone tool.

The portfolio is organized along three layers. The *skeleton layer* (Figure 3) is the discipline-neutral SMARTS thinking-partner prompt introduced in Section 4.1; any instructor can paste it into a generic LLM interface and obtain a tutor that enforces FRAME, the six-phase sequence, and the general-content/specific-application boundary. The *course-aligned layer* specializes the skeleton by adding a particular course's textbook, notation, computational stack, and topic constraints; the Course A and Course B Custom GPTs (Section 4.2) belong to this layer, as do the chapter-specific in-class assistants and the course-specific R tutor described below. The *variant layer* includes assistants that depart from the skeleton's tutoring posture for contexts that require a different interaction

contract—most notably, the Mini-Gauntlet assessment-delivery family, which refuses hints and answer reveals by default. All five families described below fall within this taxonomy; all system prompts are released in the supplementary repository (Section 8.7).

Syllabus assistants (course-aligned layer). A lightweight assistant loaded with the course syllabus answers recurring student questions about policies, deadlines, grading weights, and logistics. The knowledge base is restricted to the syllabus PDF itself, and the assistant refuses to comment on content outside that document. The practical effect is a reduction in repetitive email traffic without removing the instructor from substantive teaching interactions.

Chapter-specific in-class assistants (course-aligned layer). A set of topic-bounded assistants, one per chapter or lecture unit, supports students during and immediately after class. Each assistant is loaded only with the corresponding chapter's notes, worked examples, and R code, and declines to discuss material outside its scope. This narrowness is the design feature: it prevents the assistant from drifting into advanced territory that would confuse a student who is still consolidating the current lecture, and it gives the instructor confidence that in-class use of the assistant remains aligned with the day's learning objectives.

Course-specific R tutor (course-aligned layer). For the introductory course, a dedicated R assistant was built after observing that generic conversational AI overwhelmed statistics novices with syntax alternatives and package choices. The assistant is restricted to the `mosaic` and `BSDA` packages (the packages used in the course), explains syntax explicitly rather than assuming prior programming background, and foregrounds examples at the level of an introductory statistics course. A representative interaction is reproduced in Appendix D (Session 5). This narrowness directly operationalizes the knowledge-alignment argument developed in Section 2.2: a domain-constrained assistant avoids the off-syllabus drift that generic LLMs routinely exhibit.

Exam review and practice generators (course-aligned layer). A third family of assistants is loaded with an archive of past exams and is configured to generate practice problems that match the style, difficulty, and topic coverage of the instructor's assessments. Students can work practice problems at any hour, receive worked-solution guidance in the SMARTS style (hints privileged over answers), and self-check their readiness before sit-down exams. These assistants address a well-known asymmetry in student access to high-quality practice material: students with strong support networks have long been able to obtain tailored practice problems, while others have not.

Mini-Gauntlet assessment-delivery family (variant layer). A distinct fifth family, departing from the tutoring model entirely, delivers uniform in-class assessments directly through a Custom GPT. The assistant follows an instructor-authored blueprint specifying topic quotas and difficulty mixes, randomizes scenario details across students while holding structure and difficulty constant, enforces a two-attempt limit per item, and scores each response on both answer correctness and rationale quality. The design contains an explicit equality clause: every student receives the same blueprint with the same topic distribution and difficulty mix, so that randomization produces per-student variation without producing per-student unfairness. Unlike the tutoring assistants, this family is configured to refuse hints and answer reveals by default. A representative transcript appears in Appendix D (Session 6).

The design principle common to the entire portfolio is the same one that motivates SMARTS: narrow the assistant's knowledge, scope, and permitted behaviors so that the pedagogical intention is built into the configuration rather than relying on student self-discipline or instructor policing. We release the system prompts for all five families in the supplementary repository (Section 8.7) under a Creative Commons Attribution (CC BY 4.0) license, consistent with the license for the skeleton prompt (Figure 3); reuse and adaptation are permitted provided this paper is cited. These materials are not offered as validated interventions; they are offered as starting points from which other instructors can adapt and iterate.

8.7 Future Directions

Several extensions are warranted. First, replication at multiple institutions with larger and more diverse samples would strengthen the evidence base. Second, a controlled study comparing SMARTS-governed interactions with unconstrained LLM use (or with a no-AI condition) on both attitudinal and performance outcomes would permit causal inference. Third, the SOLO-adapted rubric should be validated through inter-rater reliability studies and applied to a larger corpus of transcripts to characterize the distribution of reasoning levels more precisely. Fourth, the relationship between transcript quality (SOLO level) and learning outcomes (examination performance) should be investigated. Fifth, the FRAME checklist, which was introduced here as a sub-tool within SMARTS Phase 1, merits independent evaluation as a prompt literacy intervention applicable across quantitative disciplines.

As LLM technology continues to evolve, the pedagogical wrapper provided by SMARTS and FRAME can be retuned to new models and platforms without abandoning the underlying design principles: keep the student as the active agent of reasoning, structure each interaction around the investigative cycle, and make the process visible and auditable.

Use of Generative AI

This manuscript is a majority-original contribution by the human authors. The following disclosure follows the structural model of Cruz and co-authors (2026) for specificity. The two Custom GPT assistants described in this paper were built on OpenAI's ChatGPT platform (ChatGPT-4o and, in later iterations, ChatGPT-5) using OpenAI's Custom GPT configuration interface; their instructions, knowledge packs, and refusal-and-escalation clauses were authored by the instructor and are released in full at the repository cited in Section 8.7.

During manuscript preparation, the authors used ChatGPT (GPT-4o and GPT-5) as a writing assistant with explicit prompt discipline intended to prevent generative rewriting. Prompts to the tool were restricted to editorial suggestions rather than content generation (for example, "Suggest editorial changes to this paragraph but do not automatically rewrite the text"), consistent with the human-in-the-loop posture advocated by this paper. Data cleaning, statistical analysis (Python 3 with the `pandas`, `numpy`, `scipy`, and `statsmodels`

libraries), table and figure preparation, and all methodological decisions were performed by the human authors. All AI-assisted editorial suggestions were reviewed and accepted or rejected by the authors, who take full responsibility for the final manuscript. Generative AI did not produce the substantive content of this paper.

Ethics Approval

Human participants research was approved by the university's Institutional Review Board (protocol number anonymized for review). Only data from consenting participants are reported.

Data Availability

De-identified item-level survey data, analysis code in Python and R, the complete 11-item survey instrument, the SOLO-adapted coding rubric with anchor examples, the full system instructions for both Custom GPT assistants, the SMARTS and FRAME reference cards, the anonymized syllabi, and reproduction scripts that regenerate every table and statistic reported in this paper are available in an anonymized public repository hosted on the Open Science Framework (OSF):

<OSF URL to be inserted in camera-ready version; anonymized preview link provided to the editor under separate cover>

The repository is structured as follows: `/code` (analysis scripts), `/data` (de-identified item-level Canvas exports), `/instrument` (survey items and SOLO rubric with anchor examples), `/prompts` (Custom GPT system instructions for both courses), `/syllabi-anonymized` (course syllabi with institution identifiers redacted), and `/reference-cards` (the SMARTS and FRAME printable handouts). A top-level `README.md` and `MANIFEST.md` describe contents and provenance.

To protect student privacy and comply with FERPA and IRB restrictions, individual-level student–LLM conversation transcripts are not publicly shared; representative excerpts appear in Section 4.3 and Appendix D.

Acknowledgments

A Survey Instrument

The 11-item attitudinal instrument was administered at three waves (T1 early semester, T2 mid-semester, T3 late semester) in both courses. Each item used a 5-point Likert-type response scale (1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree). The five open-ended items appearing at the end of each wave are also reproduced below. The full instrument is released as a plain-text file in the supplementary repository (Section 8.7).

1. The AI tutor has enhanced my overall learning experience in this course.
2. The AI tutor encourages me to think critically and solve problems independently.
3. The AI tutor helps me recall and apply previous concepts relevant to the task at hand.
4. The AI tutor is user-friendly and accessible during class activities.
5. The feedback provided by the AI tutor is as helpful as feedback from the instructor.
6. The prompts provided by the instructor help me effectively utilize the AI tutor.
7. The AI tutor supports my learning regardless of my background or prior experience with similar tools.
8. I feel prepared before attending lecture, especially when I use prompts or AI tools to guide my preparation.
9. I can understand examples covered in lecture, particularly those that incorporate AI tools or resources.
10. The course materials and AI tutor reduce my need to seek additional resources beyond what is provided by the instructor.
11. I am aware of the university's efforts to integrate AI into the classroom.

Open-ended prompts (administered alongside the Likert items).

1. Considering your work–life–class balance, how do AI tools (e.g., ChatGPT, Bard) help or hinder your ability to manage and succeed?
2. What features of the AI tutor did you find most helpful? What features need improvement? Please provide specific examples.
3. Do you feel the AI tutor accommodates the needs of all students equally? If not, what specific challenges or barriers have you or others experienced?
4. How has using the AI tutor affected your ability to think critically and solve problems independently? Can you describe a specific instance where it helped or hindered you?
5. Any suggestions to improve these AI tutoring systems?

Full 11-item wave-by-wave descriptive and trend results are provided in Table 6.

Table 6: Full item-level paired Likert results (T1 → T3) for the 11-item student-attitude instrument, by course. Wilcoxon signed-rank test (exact, Pratt zero handling) with matched-pairs rank-biserial correlation r_{rb}. Reproduction code in the supplementary repository.

Course	Item	Description	n	T1 M (SD)	T3 M (SD)	$\bar{\Delta}$	+/0/−	V	p	r_{rb} (95% CI)
A	Q1	AI tutor enhances learning	13	4.00 (0.58)	4.08 (0.76)	+0.08	5/5/3	31.0	1.000	+0.15 [−0.41, +0.69]
A	Q2	Encourages critical thinking	13	3.54 (0.78)	3.77 (0.83)	+0.23	6/5/2	22.0	0.588	+0.35 [−0.21, +0.89]
A	Q3	Helps recall/apply prior concepts	13	3.92 (0.49)	4.31 (0.63)	+0.38	4/9/0	0.0	1.000	+0.51 [+0.14, +0.77]
A	Q4	User-friendly and accessible	13	4.00 (0.71)	4.31 (0.63)	+0.31	3/10/0	0.0	0.542	+0.40 [+0.00, +0.69]
A	Q5	Feedback as helpful as instructor	13	3.69 (0.95)	4.08 (0.86)	+0.38	5/7/1	10.0	0.635	+0.47 [+0.00, +0.84]
A	Q6	Instructor prompts help use of AI tutor	13	3.85 (0.80)	4.23 (0.60)	+0.38	5/7/1	10.0	0.635	+0.47 [+0.00, +0.84]
A	Q7	Supports learning regardless of background	13	4.08 (0.49)	3.77 (0.93)	−0.31	2/7/4	19.0	0.068	−0.27 [−0.71, +0.27]
A	Q8	Feel prepared before lecture	13	3.38 (0.87)	4.08 (0.76)	+0.69	8/4/1	8.5	0.068	+0.70 [+0.24, +0.97]
A	Q9	Can understand examples in lecture	13	3.69 (0.75)	4.23 (0.60)	+0.54	5/8/0	0.0	0.542	+0.60 [+0.27, +0.89]
A	Q10	Reduces need for additional resources	13	3.85 (0.69)	4.15 (0.55)	+0.31	5/7/1	10.5	0.685	+0.46 [+0.00, +0.84]
A	Q11	Aware of university AI integration efforts	13	3.77 (0.60)	4.00 (0.41)	+0.23	3/10/0	0.0	0.542	+0.40 [+0.00, +0.69]
B	Q1	AI tutor enhances learning	16	3.88 (0.62)	4.00 (0.52)	+0.12	6/7/3	39.0	0.980	+0.22 [−0.31, +0.73]
B	Q2	Encourages critical thinking	16	3.62 (0.50)	3.69 (0.60)	+0.06	4/9/3	39.0	0.433	+0.10 [−0.40, +0.55]
B	Q3	Helps recall/apply prior concepts	16	4.19 (0.75)	4.12 (0.72)	−0.06	2/11/3	28.0	0.039	−0.10 [−0.51, +0.33]
B	Q4	User-friendly and accessible	16	4.19 (0.75)	4.38 (0.62)	+0.19	6/7/3	36.0	0.860	+0.26 [−0.24, +0.73]
B	Q5	Feedback as helpful as instructor	16	3.62 (0.81)	3.50 (0.97)	−0.12	3/8/5	38.5	0.144	−0.17 [−0.64, +0.33]
B	Q6	Instructor prompts help use of AI tutor	16	3.94 (0.77)	4.00 (0.73)	+0.06	7/4/5	56.0	0.940	+0.10 [−0.43, +0.62]
B	Q7	Supports learning regardless of background	16	4.00 (0.97)	4.12 (0.62)	+0.12	4/9/3	37.5	0.495	+0.12 [−0.37, +0.60]
B	Q8	Feel prepared before lecture	16	3.25 (0.93)	3.69 (0.87)	+0.44	6/5/5	45.0	0.706	+0.23 [−0.34, +0.69]
B	Q9	Can understand examples in lecture	16	3.81 (0.91)	3.75 (0.77)	−0.06	5/6/5	55.0	0.528	−0.04 [−0.55, +0.48]
B	Q10	Reduces need for additional resources	16	3.69 (0.79)	3.50 (0.63)	−0.19	3/7/6	39.0	0.144	−0.22 [−0.71, +0.30]
B	Q11	Aware of university AI integration efforts	16	4.38 (0.72)	4.44 (0.51)	+0.06	3/12/1	16.0	0.193	+0.19 [−0.23, +0.60]

Note. Paired analysis: n is the number of students with complete T1 and T3 responses on the item. $\bar{\Delta} = T3 - T1$. +/0/− = number of students whose T3 response exceeded, matched, or fell below their T1 response. V = Wilcoxon signed-rank statistic (sum of positive signed ranks; Pratt zero handling). p = exact two-sided p-value. r_{rb} = matched-pairs rank-biserial correlation (Kerby, 2014); 95% CI by 5,000-replication percentile bootstrap. All figures reproducible via supplementary code (`run_paired_main.py`).

B Sensitivity Analyses

This appendix presents the three-wave repeated-measures and parametric sensitivity analyses referenced in Section 6.5. All three analyses were conducted independently from the primary T1 $\rightarrow$ T3 paired Wilcoxon analysis by a second analyst as a robustness check. Reproduction code and raw output are in the supplementary repository (Section 8.7).

Friedman omnibus and Page's L ordered-trend tests

Both tests require complete observations at all three waves, so they are computed on the strict complete-case subsample ($n = 11$ in Course A, $n = 14$ in Course B; Figure 5). Friedman's test is the omnibus repeated-measures analog of Kruskal–Wallis under within-subject dependence (Hollander et al., 2014); Page's L tests for a monotone trend across the ordered levels T1 $\leq$ T2 $\leq$ T3 and is the paired analog of the Jonckheere–Terpstra trend test (Page, 1963; Hollander et al., 2014). Exact p-values are reported (by permutation for Page's L, by asymptotic χ^2 approximation for Friedman).

Table 7: Friedman omnibus and Page's L ordered-trend tests on the three-wave complete-case subsample. Course A: $n = 11$. Course B: $n = 14$.

Course	Item	Description	Friedman χ^2	p	Page's L	p (exact)
A	Q1	AI tutor enhances learning	0.23	0.891	132.0	0.542
A	Q2	Encourages critical thinking	1.03	0.597	136.0	0.231
A	Q3	Helps recall/apply prior concepts	3.00	0.223	135.0	0.300
A	Q4	User-friendly and accessible	0.74	0.692	134.0	0.376
A	Q5	Feedback as helpful as instructor	2.33	0.311	136.5	0.231
A	Q6	Instructor prompts help use	1.60	0.449	135.0	0.300
A	Q7	Supports learning regardless of background	1.46	0.483	128.0	0.829
A	Q8	Feel prepared before lecture	5.36	0.069	139.5	0.084
A	Q9	Can understand examples in lecture	4.00	0.135	138.0	0.122
A	Q10	Reduces need for additional resources	2.33	0.311	135.0	0.300
A	Q11	Aware of university AI integration efforts	3.13	0.210	134.5	0.376
B	Q1	AI tutor enhances learning	2.53	0.282	172.0	0.256
B	Q2	Encourages critical thinking	2.25	0.325	168.0	0.537
B	Q3	Helps recall/apply prior concepts	1.33	0.513	168.0	0.537
B	Q4	User-friendly and accessible	2.21	0.331	171.5	0.320
B	Q5	Feedback as helpful as instructor	1.54	0.462	168.0	0.537
B	Q6	Instructor prompts help use	0.63	0.729	171.0	0.320
B	Q7	Supports learning regardless of background	1.80	0.407	168.0	0.537
B	Q8	Feel prepared before	2.80	0.247	175.0	0.111

Pooled ordered logistic regression

As a parametric sensitivity cross-check, we fit an ordered logistic regression for each item $\times$ course on the full unbalanced panel (Course A: 39 wave-level observations across 14 students; Course B: 48 observations across 18 students) with wave entered as an ordered numeric predictor $t \in \{1, 2, 3\}$ (Agresti, 2010). The specification does not model within-subject correlation (the $n = 11/14$ complete-case subsample is at the lower edge of the Bolker et al. recommendation for mixed ordinal models; Bolker et al., 2009), so this regression is sensitivity rather than primary.

Table 8: Pooled ordered logistic regression of item-level response on wave (time as ordered numeric, $t \in \{1, 2, 3\}$) on all wave-level observations (Course A: $n_{\text{obs}} = 39$ across 14 students; Course B: $n_{\text{obs}} = 48$ across 18 students).

Course	Item	Description	β	OR	p	95% CI (OR)
A	Q1	AI tutor enhances learning	+0.10	1.11	0.779	[0.54, 2.27]
A	Q2	Encourages critical thinking	+0.25	1.28	0.491	[0.63, 2.61]
A	Q3	Helps recall/apply prior concepts	+0.53	1.70	0.158	[0.82, 3.52]
A	Q4	User-friendly and accessible	+0.41	1.50	0.268	[0.73, 3.07]
A	Q5	Feedback as helpful as instructor	+0.37	1.44	0.303	[0.72, 2.90]
A	Q6	Instructor prompts help use	+0.41	1.51	0.263	[0.73, 3.08]
A	Q7	Supports learning regardless of background	−0.34	0.72	0.369	[0.34, 1.49]
A	Q8	Feel prepared before lecture	+0.63	1.87	0.087	[0.91, 3.85]
A	Q9	Can understand examples in lecture	+0.71	2.04	0.076	[0.93, 4.50]
A	Q10	Reduces need for additional resources	+0.39	1.47	0.287	[0.72, 2.99]
A	Q11	Aware of university AI integration efforts	+0.23	1.26	0.500	[0.64, 2.48]
B	Q1	AI tutor enhances learning	+0.31	1.36	0.384	[0.68, 2.73]
B	Q2	Encourages critical thinking	+0.32	1.37	0.395	[0.66, 2.85]
B	Q3	Helps recall/apply prior concepts	−0.10	0.91	0.769	[0.47, 1.75]
B	Q4	User-friendly and accessible	+0.31	1.37	0.347	[0.71, 2.61]
B	Q5	Feedback as helpful as instructor	−0.15	0.86	0.635	[0.46, 1.60]
B	Q6	Instructor prompts help use	+0.09	1.09	0.785	[0.58, 2.08]
B	Q7	Supports learning regardless of background	+0.10	1.10	0.777	[0.57, 2.14]

Summary of sensitivity findings

All three sensitivity analyses (Friedman, Page's L, pooled ordered logistic) are directionally concordant with the primary paired T1 $\rightarrow$ T3 Wilcoxon analyses:

1. The three Course A items with rank-biserial CIs excluding zero in the primary analysis (Q3, Q8, Q9) also produce the smallest Friedman and Page's L p-values in the three-wave sensitivity.

2. No item reaches $p < 0.05$ under any sensitivity test, just as no item reaches $p < 0.05$ under the primary exact paired Wilcoxon test. The effect-size-first decision rule that weights compatibility intervals above p-value thresholds (Section 6.5) is preserved across all four specifications.

3. The sign concordance between ordered-logit β and paired rank-biserial r_{rb} on 21 of 22 item $\times$ course combinations constitutes a parametric cross-check: agreement does not prove independent corroboration but it rules out large-effect artifacts of the non-parametric specification.

The paper's substantive interpretation does not change under any of these specifications. The primary T1 $\rightarrow$ T3 paired analysis is retained as the main-text specification because it uses the larger subsample ($n = 13$ and $n = 16$ vs. $n = 11$ and $n = 14$), directly addresses the central research question of start-to-end shift, and matches the pre-specified effect-size-first decision rule. The sensitivity analyses are reported here and in the supplementary repository for full transparency.

C Anonymized Course Syllabi

The syllabi for Course A (Introductory Statistics) and Course B (Advanced Mathematical Statistics) are reproduced in the supplementary repository with all institution-identifying strings (university name, logos, instructor contact information, physical room numbers) redacted. In-text course numbers and references to the institution in this manuscript have likewise been replaced by generic descriptors for blind review.

D Annotated SMARTS Sessions Across the Course-Aligned Assistant Portfolio

This appendix reproduces four complete student–LLM transcripts underlying the annotated excerpts in Section 4.3 (Sessions 1 through 4). Two additional sessions (Sessions 5 and 6) are summarized briefly below and released in full in the supplementary repository; they are included in this portfolio specification because they illustrate two further modes of SMARTS deployment—the R-tutor custom GPT in routine lab use (Course A) and the Mini-Gauntlet assessment-delivery assistant under an attempted circumvention (Course B)—that the four main-appendix transcripts do not cover on their own. SMARTS phase labels are added by the coder (the instructor) on Sessions 1 through 4; a SOLO-rubric classification appears on the student's highest-level turn. Transcripts are reproduced in full except for (i) redaction of any student identifying information and (ii) light formatting adjustments for the journal typesetter (line wrapping, consistent notation for $\hat{\theta}$ and $Y_{(1)}$, removal of PDF export footers from the source export). No statistical or procedural content has been altered from what the student and assistant produced in real time.

Of the six sessions, three use the Course-A custom GPTs (Sessions 1, 2, 5), two use the Course-B custom GPTs (Sessions 3 and 6), and one uses a student-written system prompt (Session 4). The pairing of Session 3 with Session 4 supports the direct free-prompt-versus-custom-GPT comparison developed in Section 4.3. The Mini-Gauntlet session (Session 6) illustrates the refusal-by-default posture used for assessment delivery, as distinct from the coaching posture in the other sessions (Section 8.6).

The full verbatim PDF exports of all six transcripts are released in the supplementary repository (Section 8.7).

Session 1: Course A — Hypothesis test on a manufacturer's efficacy claim (custom GPT)

Student opening: "can u do this problem for me"

The assistant declines and routes the student to a FRAME decomposition. The full transcript covers approximately twelve turns across the six SMARTS phases and closes with

the student producing a complete hypothesis-test argument against the manufacturer's claim. The annotated excerpt in Section 4.3 (Example 1) reproduces the pivotal Phase S, M, R, and summary turns. The full reproduction below preserves the natural student-turn sequencing.

[**Full transcript reproduced in supplementary repository; annotated excerpt in Section 4.3, Example 1.**]

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates test statistic, degrees of freedom, one-sided p-value, significance decision, and the context of the manufacturer's claim into a coherent inferential statement.

Session 2: Course A — One-way ANOVA with Tukey HSD post-hoc on GPA data (custom GPT)

Student opening: "So basically I have a question to compare four universities [...]"

The assistant opens with explicit SMARTS phase labels ("Phase S — State and clarify," etc.), requires a complete FRAME before proceeding, and iterates through three rounds of FRAME tightening before accepting the student's opening. The transcript then moves through R code debugging (an initial `aov()` formula error), the full ANOVA analysis, a refusal to move to the Tukey post-hoc until the student articulates why the post-hoc is needed, a Tukey interpretation error catch, and an explicit refusal to close on a one-word "ty" response. The student ultimately produces an ordered conclusion with the correct directionality.

[**Full transcript reproduced in supplementary repository; annotated excerpt in Section 4.3, Example 2.**]

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the ANOVA omnibus result, the family-wise error rationale for the post-hoc test, the correct Tukey decision rule applied across all six pairs, and the ordered conclusion into a coherent claim about the four groups.

Session 3: Course B — Unbiasedness of $\hat{\theta} = nY_{(1)}$ with an exponential sample (custom GPT with explicit boundary)

Student opening: "so for part 1 im struggling, its asking to show theta hat is unbiased? first can you tell me what does it mean to be an unbiased estimator?"

The "Proof Construction Coaching" custom GPT opens with an explicit statement of its scope boundary: "I'll share general theorems, definitions, and standard formulas freely, but I won't do the application to your specific problem. That part is your proof." The assistant supplies the definition of unbiasedness and the general minimum-density formula $f_{Y_{(1)}}(y) = n(1 - F(y))^{n-1} f(y)$, but stops the student when they attempt to jump to the specific exponential substitution $= ne^{-ny/\theta}$. The assistant routes the student back to explicit identification of $F(y)$, $1 - F(y)$, and $f(y)$, catches a missing $1/\theta$ factor by forcing the student to self-diagnose, requires the student to match both coefficient and exponent when recognizing the exponential form, and closes with a structural pipeline summary (general formula $\rightarrow$ substitute $\rightarrow$ recognize distribution $\rightarrow$ apply standard expectation $\rightarrow$ apply linearity) rather than a completed derivation.

[**Full transcript reproduced in supplementary repository; annotated excerpt in Section 4.3, Example 3.**]

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the order-statistic minimum-density formula, the specific exponential substitution, recognition of the exponential rate parameter, the standard $E[Y] = 1/\lambda$ result, and linearity of expectation into a complete unbiasedness proof.

Session 4: Course B — Unbiasedness of S^2 with a student-written system prompt

Student opening (system prompt, verbatim from the transcript): "I want you to adopt the persona of a supportive tutor who encourages a step-by-step approach to solving statistics problems. ... validate correct reasoning, clarify misconceptions, and guide me toward the next step. If I struggle, ask whether I'd like a hint or an example. If I request the entire solution, gently encourage a step-by-step effort. ... keep your guiding statements few

sentences and hints, not entire solution. ... When all else fails, reveal only the first step and prompt me to proceed."

The student's self-written prompt is pedagogically reasonable and shares structural features with the SMARTS specification (step-by-step emphasis, hints over full solutions, first-step revelation as a fallback). It differs from the SMARTS custom GPTs in that it does not enforce a FRAME opener, it allows the assistant to complete identities after explicit student consent, and it privileges confirmation ("you're spot on," "exactly right") alongside the hint-privileging. The transcript covers the unbiasedness of S^2 with $E[Y_i] = \mu$ and $\mathrm{Var}(Y_i) = \sigma^2$. The student reaches the relational level by using the variance-decomposition identity and the $E[X^2] = \mathrm{Var}(X) + (E[X])^2$ relation.

[**Full transcript reproduced in supplementary repository; annotated excerpt in Section 4.3, Example 4.**]

SOLO level of the final student turn: Relational. *Rubric justification:* the student integrates the variance-decomposition identity, the variance-mean relation $E[X^2] = \mathrm{Var}(X) + (E[X])^2$, linearity of expectation, and the $\mathrm{Var}(\bar{Y}) = \sigma^2/n$ result into a coherent unbiasedness argument.

Session 5: Course A — R-tutor custom GPT in routine use (histogram construction and interpretation)

This session illustrates the R-tutor custom GPT in a routine lab-assignment context rather than in the exam-review or homework-coaching contexts of Sessions 1 and 2. The R-tutor is restricted to the `mosaic` and `BSDA` packages used in Course A and is configured to explain `R` syntax explicitly rather than assuming prior programming background. The transcript covers the construction, interpretation, and revision of a histogram for a skewed dataset, with the assistant coaching the student through bin-width choice and the decision to present a log-transformed axis. The complete transcript and its SOLO-rubric annotation are released in the supplementary repository (Section 8.7, folder `transcripts/session05/`), which also includes the R-tutor's full system prompt and the raw lab dataset.

Session 6: Course B — Mini-Gauntlet assessment-delivery assistant (attempted circumvention of the refusal-by-default posture)

This session illustrates the Mini-Gauntlet assistant family described in Section 8.6, whose system prompt is configured to refuse hints and answer reveals by default and to enforce a two-attempt cap on each item. The transcript is included specifically to show the assistant's behavior under an attempted circumvention: the student asks the assistant to reveal the answer after an incorrect first attempt, the assistant declines and restates the two-attempt policy, the student tries a reframing request, and the assistant continues to refuse while offering rationale feedback. This session therefore illustrates a different mode of the SMARTS design than the coaching examples: the refusal-by-default posture appropriate to assessment delivery rather than tutoring. The complete transcript is released in the supplementary repository (Section 8.7, folder `transcripts/session06/`), together with the Mini-Gauntlet system prompt and the item set used in the assessment.

Cross-session comparison

All six sessions illustrate different modes of the SMARTS design but share a common posture at the inflection points of the reasoning. Sessions 1 through 3 (Lipitor, ANOVA/Tukey, unbiasedness of $\hat{\theta} = nY_{(1)}$) display a pattern of refusal-and-redirect at the moment a student attempts to skip a reasoning step: the Lipitor session refuses to produce the p-value; the ANOVA session refuses to advance to the Tukey post-hoc until the student articulates family-wise error; the unbiasedness session refuses to confirm the specific exponential substitution. Session 4 instead confirms each step as it is produced, reaching the same relational SOLO level on a student with strong foundations but with softer scaffolding. Session 5 shows the R-tutor custom GPT operating within a narrow computational domain (`mosaic/BSDA`) rather than across a full problem arc. Session 6 shows the Mini-Gauntlet assessment-delivery posture, where the refusal mechanism is the *whole* contract rather than a periodic intervention.

As noted in Section 6.6, this appendix is illustrative rather than a statistical demonstration that the SMARTS scaffolding is necessary for the relational outcome: six transcripts cannot support such a claim, and the observed relational-level outcome in Session 4

confirms that a student with adequate foundations can reach that level through a softer scaffolding regime. What the appendix does support is the concrete specification of the SMARTS scaffolding across its coaching, computational, and assessment-delivery modes, against which future comparative studies with paired student populations and randomized scaffold assignment can be designed.

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